

2020 K-Innovation ODA Program with Lao PDR

National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022

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Title	National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022
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National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022

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Executive Summary

The government of Lao PDR plans to have the country graduate from being the Least Developed Country (LDC) and focuses on the significance of Science, Technology, and Innovation as a tool for economic growth and national development plan. The National Master Plan for Technology and Innovation Development and Application in Lao PDR (NMTIDL) and National Master Plan for Policy on Science and Technology Research (NMPSTR) are both going to be the priority project that will be included in the national development plan of science and technology in 2019-2025.

With the goal of formulating a master plan and enhancing support for the development of the 8th National Socio-Economic Development Plan and contributing to the preparation of the 9th National Socio-Economic Development Plan, Lao PDR's Development Strategy and Vision up to year 2030 has identified the National Science and Technology Development Program as a driving force for economic growth. To be sustainable, green, and consistent with the SMEs Development Strategies, the national development plan needs to be associated with the transformation of SMEs for the old business production going to the new modern economics of science, technology, and innovation.

The objective of the K-Innovation ODA Project with Lao PDR (2020-2022) is to set up the first National Master Plan for STI R&D and Application in Lao PDR and Master plan on the Application of Technology and Innovation for Sectoral Support for National Socio-Economic Development Growth in the future as well as Sectors under development. Moreover, the project aims to strengthen the capacity of key institutions of the Ministry of Science and Technology (MOST) in the mid- and long-term strategic planning of STI in Lao PDR.

Based on the confirmation and acceptance of the project proposal submitted by MOST, the Korean government affirmed support for Lao PDR in making faster progress toward achieving Lao PDR's STI development goals through the K-Innovation ODA program "National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022."

The scope of the Project has been defined based on Lao PDR's STI policy priorities and the lessons learned from the Korean experiences in national STI policies and technology foresight. Accordingly, the Project framework includes the following: i) review of Vietnam's national STI strategies and technology foresight experiences; ii) research on the Korean experiences in technology foresight and national STI strategies; iii) local workshops for the diagnosis of Vietnam's technology foresight capacities and for the consensus on the policy consultation's direction and contents; iv) capacity-building workshop with technical training programs to enhance the technology foresight capacity of Vietnamese government officials and experts; and v) development of implementation plan for technology foresight by VISTI and review process by STEPI and Korean experts.

At the kick-off meeting with MOST, the objectives and scope of the Project were discussed. STEPI and MOST discussed further specific action plans and agreed on the project programs and activities. The research team suggested implications and recommendations on how to develop Lao PDR's capacity for national STI strategy. Regarding the national STI strategy, it is necessary to make the government establish and implement an STI strategy systematically with active commitment to making STI contribute to national socio-economic development. Lao PDR needs to promote strategic investments and systematic management to achieve the best possible outcomes given the limited R&D resources. Lao PDR should also improve its R&D structure to increase the performance of the R&D system considering the limited R&D resources. For this, the role of R&D organizations should be clearly identified.

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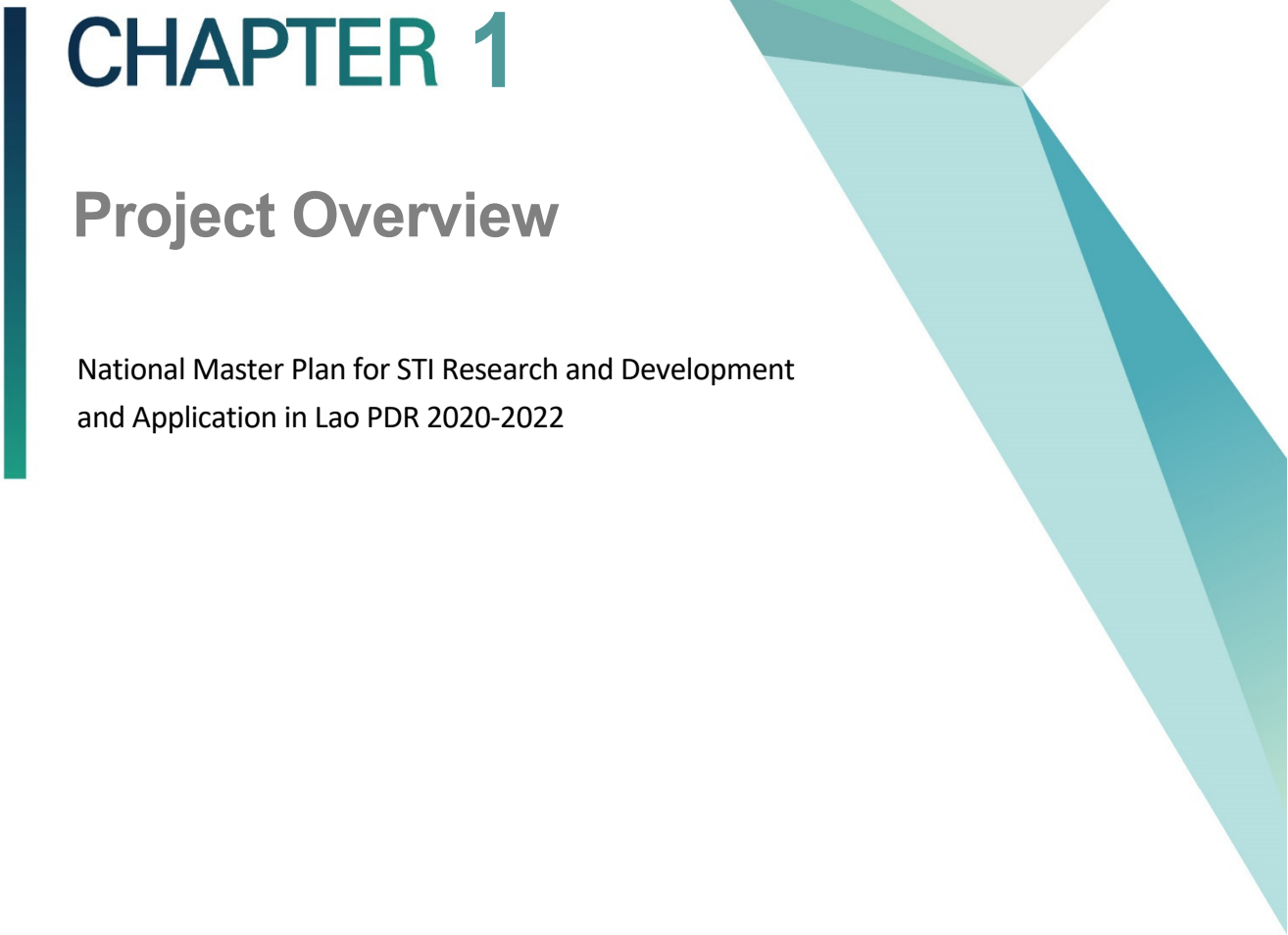
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CHAPTER 1

Project Overview

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022



Chapter 1. Project Overview

1. Introduction

The government of Lao PDR plans to have the country graduate from one being the Least Developed Country (LDC) and focuses on the significance of Science, Technology, and Innovation as a tool for economic growth and national development plan. The National Master Plan for Technology and Innovation Development and Application in Lao PDR (NMTIDL) and National Master Plan for Policy on Science and Technology Research (NMPSTR) are both going to be the priority project that will be included in the national development plan of science and technology in 2019-2025.

With the goal of formulating a master plan and enhancing support for the development of the 8th National Socio-Economic Development Plan and contributing to the preparation of the 9th National Socio-Economic Development Plan, which will be followed in the coming years, and the country's Development Strategy and Vision up to year 2030 wherein the National Science and Technology Development Program has been identified as a driving force for economic growth to be sustainable, green, and consistent with the SMEs Development Strategies associated with the transformation of SMEs for the old business production going to the new modern economics of science, technology, and innovation.

The Second Master Plan will reach the goal, but this project really requires support from various sectors such as government, private sector, research institutions, banking, and public sector and international cooperation and support, particularly the assistance of the government of the Republic of Korea such as STEPI in the form of technical consultation, as well as some activity budgeting and other forms of assistance.

To see the results of implementation of National Socio-Economic Development in 5 years, period VIII, national strategy on science and technology development, as the national master plan for technology and innovation development and application, is necessary and important to encourage and promote the implementation of sectoral strategies through studying and selection of advanced and appropriate technologies. To determine the project on technology development and application for the priority sector, R & D for innovation products and research on science and technology are a major task to promote National Socio-Economic Development and Growth. Thus, the government has designated the research and development of science and technology as the main driving force of commercial productivity and production, which has affected the changing economic structure of sustainable growth, aiming to achieve the development goal of graduating from being the least developed country in 2020 to one of the middle-income developing countries. Green growth and sustainability in 2030 are stated in the vision until 2030 and the 10-year socio-economic development strategy (2016-2025).

In order to ensure the achievement of this goal, the National Master Plan for technology and innovation development and application in Lao PDR and National Policy on Science and Technology Research must be developed, with the aim of focusing development on the science and technology research master plan, survey and study development indicator of the relevant sectors, human resources, science and technology research innovation networking, and infrastructure development such as research facilities as well as building a center for science and technology research transfer knowledge for increasing trading, research competitiveness, and industrial development to become more innovative.

The Department of Technology and Innovation (DTI) acts as a secretariat to the Ministry of Science and Technology to formulate policies and strategies and manage and promote the transfer technology and innovation of Lao PDR. The Cabinet of Academy of Science and Technology (CAST) acts as a secretariat to the Ministry of Science and Technology and

National Science Council (NSC) in order to manage the research on science and technology and connection to the science sectors and related organization within the central and local governments to implement and certify S&T research projects as well as set up the National Master plan for R&D and technology application for sectoral support of Lao PDR.

The Science and Technology Policy Institute (STEPI) is a Korean government-affiliated research institute under the Prime Minister's Office and a leading global think tank that conducts research and analysis on issues pertaining to science, technology, and innovation. STEPI works closely with various ministries, regional governments, and relevant institutes to support policy design. It also has a long experience in providing STI policy consultancy for developing countries, having the role of conducting STI development cooperation activities with partner countries.

2. Objectives

The project is titled “National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022.” The program duration is from January 1, 2020 to December 31, 2020 (12 months).

1) The research objective of the K-Innovation ODA Project with Lao PDR (2020-2022) is to set up the first National Master Plan for STI R&D and Application in Lao PDR and Master Plan on the Application of Technology and Innovation for Sectoral Support for National Social-Economic Development Growth in the future as well as Sectors under development. Moreover, the project aims to strengthen the capacity of key institutions of the Ministry of Science and Technology in the mid- and long-term strategic planning of STI in Lao PDR.

2) The detailed objectives are as follows:

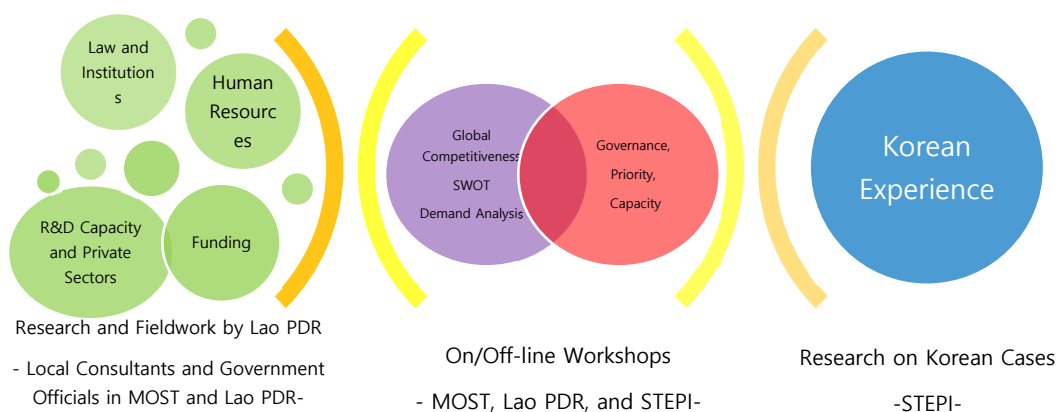
- Review the current status of STI policy and infrastructure in Lao PDR
- Strengthen the capacity building of policymakers, developers and scientists, and young researchers and identify the priority areas of research and development of the National STI Research Program
- Share Korea’s experience in developing the master plan for national STI
- Conduct a capacity-building workshop
- Provide technical advice on the draft report of the master plan for STI
- Provide a roadmap for establishing the National R&D Institute

3. Project Framework

The Project's main activities will be framed as follows:

- Review of the current status of Lao PDR's national STI and related infrastructure
- Survey and study development indicator of the relevant sector
- Research on the Korean experience in the establishment and development of national STI master plan and strategies or plan
- Online/Offline workshops for Lao PDR to enhance knowledge and capacities and to reach consensus on the direction and contents of policy consultation for major policymakers, government officials, and research institutes of Lao PDR
- Capacity-building workshop to enhance the technology foresight capacity of Lao PDR's government officials and experts in the STI field through technical training programs
- Draft the National Master Plan for STI and STI policy development in Lao PDR and review process by STEPI and Korean experts

[Figure 1-1] Conceptual Framework of the Project



Source: author

4. Project Team

4.1 Research Team

The research team is composed of three full-time STEPI researchers and three experts from other institutes. The experts are invited to share their knowledge of Korean development experiences.

The research team from STEPI consists of three researchers, including Dr. Ji Hyun Kim as Project Manager (PM) and two STI policy experts in STEPI. The research team is responsible for the research and review on Lao PDR's STI. Three other experts will review and advise on STI policy strategies and share the Korean experience that can be adopted for Lao PDR's STI master plan.

[Table 1-1] Research Team

Name	Organization	Area of expertise	Contact information
Ji Hyun Kim	STEPI, Associate Research Fellow Project Manager (PM)	STI Policy, Technology Foresight	jihyunkim@stepi.re.kr
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Chung Ho Lee	Institute of Ministry of Science and Technology Management, Senior Adviser	Local Mentoring	Choongholee @naver.com

4.2 Local Research Team

As the main counterpart for this project, the Ministry of Science and Technology (MOST) of Lao PDR is responsible for developing the national master plan on STI in Lao PDR. The local research team will work in collaboration with the STEPI research team.

[Table 1-2] Local Research Team

Name	Office	Position	Contact information
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5. Project Activities

5.1 Activity 1: Preparation of the Project

From January to March 2020, the STEPI team will communicate with the Ministry of Science and Technology (MOST) Lao PDR since it is the first year to initiate the ODA project with Lao PDR. STEPI and MOST will have an introduction and share the research plan based on the Project Concept Papers written in 2019 through emails. Due to the unpredictability caused by COVID-19 in February, the project plan and schedule for the year had been affected drastically, and the team needed to wait to see the possibility of visiting Lao PDR. In the meantime, STEPI drafted a contingency plan for the projects and shared it with its counterpart. The teams decided to move the kick-off meeting to May since the government officials in Lao PDR started remote working at home and it was difficult to host the online workshop due to unstable Internet connection.

5.2 Activity 2: Outlining the Work Scope

The STEPI and MOST teams outlined the work scope of the project under the COVID-19 situation and the contingency plan with the changed schedule and plan, and the list of needed data from Lao PDR was drafted and shared with MOST for coordination. Since traveling to Lao PDR or to the Republic of Korea was forbidden for several months, the research teams in STEPI and MOST started literature review and research on the current status of STI, STI policy and education, its governance and legislative infrastructure, etc. to understand the current situation and status of Lao PDR's STI policy status. STEPI reached out to hire local consultants from Lao PDR to research on baseline data and information on STI and related information in Lao PDR. The work scope was divided into three parts with the needed information for contracting with three local consultants.

5.3 Activity 3: Kick-off Meeting

The online kick-off meeting was held on June 17, 2020 through the government online meeting system link sent by MOST of Lao PDR, followed by introduction of both sides, project plan and expected activities with schedule, contracts to be made for the consultants, implementation plans, and budgets. The team exchanged Q&As regarding the project. Even until June, traveling to other countries were restricted, and the role of local consultants was critical to gather the required information and data. The need for baseline data and information on STI in Lao PDR was highlighted throughout the meeting.

5.4 Activity 4: Comments on Progress

Gathering the STI information and data in Lao PDR is the most critical part of the research to understand and develop strategies based on the facts in Lao PDR. The exchange of findings and information will be critical to understanding the current situation of STI in Lao PDR, which will be the baseline information for developing the national master plan.

5.5 Activity 5: Online Workshop

An online expert workshop will be held in early December 2020. During the workshop, STEPI and Korean experts will give a presentation on the methodology to develop a national master plan on STI strategies and share experiences in the identified topics, including national strategic planning for STI. At the same time, experts and government officials from Lao PDR will also share the current status of their STI and STI policy and demands in Lao PDR in order to enhance in-depth understanding of Lao PDR's current status and experiences. Afterward, STEPI and MOST will establish consensus on the expected results and outcome. STEPI will present the results and achievement of the cooperative project with Lao PDR and follow-up plan for 2021.

6. Project Schedule

The following is the overall schedule of the project:

[Table 1-3] Project Schedule

Activities	Jan. '20	Feb. '20	Mar. '20	Apr. '20	May '20	Jun. '20	Jul. '20	Aug. '20	Sep. '20	Oct. '20	Nov. '20	Dec. '20
Project Identification												
Literature Review and Research												
Kick-off Meeting and Workshop												
Interim Report												
Comments on Progress												
Online Workshop												
Final Report												

7. Project Outputs

STEPI and MOST will draw up the following deliverables:

- **Deliverable 1:** Project Proposal
- **Deliverable 2:** Program Agenda for Kick-off Meeting
- **Deliverable 3:** Presentations and Training Materials for Online Workshop (STEPI)
- **Deliverable 4:** Draft of report on Lao PDR's STI data and information
- **Deliverable 5:** Final Report (2020)



CHAPTER 2

National STI Policy and STI Education in Lao PDR

National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022

Chapter 2. National STI Policy and STI Education in Lao PDR

1. Lao PDR's National Development Strategies

1.1 Overview¹

1.1.1 Political Situation

After gaining its independence from France in 1945, Lao PDR went through the Indochina War and officially won its independence through the Geneva Agreement in 1954. Later, with the support of Vietnam, the socialist “Lao People's Democratic Republic” was established in 1975. Lao PDR maintains a one-party dictatorship with the Lao People's Revolutionary Party at the center and a collective leadership system wherein key government posts, including the president, head of state, prime minister, and speaker of the National Assembly concurrently serve as the top political bureau in charge of decision making of the party's central committee.

The central government has an indirectly elected president as its head of state with a five-year National Assembly resolution, with the current Bounnhang Vorachith (Bunyang Vorachit) elected in 2016 serving his term. The prime minister is appointed by the president with the approval of the National Assembly, and 18 ministries including the Ministry of National Defense, Ministry of Public Security, Ministry of Interior and Safety, Ministry of Agriculture and Forestry, Ministry of Justice, Ministry of Labor and Social Welfare, Ministry of Energy and Mining, Ministry of Commerce and Industry, Ministry of Science and Technology, Ministry of Education and Sports, Ministry of Public Affairs, Ministry of Planning,

¹ Source: JST (2014); CSIRO (2019); World Bank in Vietnam, <https://www.worldbank.org/en/country/vietnam/overview#1>, accessed 17 September 2019.

Ministry of Health, etc.

The ministries play a directing and supervising role. There is a local government in each of the 17 states and one special city (Vientiane), with the governor and the mayor of Vientiane appointed by the president upon the prime minister's recommendation operating a central government-centered administrative system.

In addition to the Central Committee, the Lao People's Revolutionary Party (LPRP) has local committees by state and lower administrative units; the national convention is held once every five years, and the Party Central Committee, twice a year, to play a role in determining the Party's guiding policy, the government's basic policy direction, etc. Serving as a supporting organization of LPRP is the Lao Front for National Development (LFND), and activities such as forming a cooperative for local residents, promoting mental reform and improving living for rural development, and forming a multi-ethnic cooperative are carried out.

Lao PDR basically maintains the legal system of the French colonial rule (1899-1954); due to the unstable political situation such as civil war and regime change even after its independence from France, however, the publication of the official gazette began in 1993, continuing into the 2000s.

The Constitution was first amended by the 5th National Assembly in April 2003 after it was enacted and promulgated on August 15, 1991. In addition to the Constitution, 143 laws are in effect (as of July 2019), with the Lao government maintaining a system for determining national norms under the one-party dictatorship of the Lao People's Revolutionary Party, proclaiming all legislative matters as presidential decrees instead of enacting detailed laws.

1.1.2 Economic Situation

Lao PDR is an inland country bordering China, Thailand, Vietnam, Myanmar, and Cambodia, with Mekong River at its center. After gaining independence from France, the socialist system was established in 1975 with the support of Vietnam. With the continuation of the

socialist system, Lao PDR has long remained one of the poorest countries with stagnant growth; to overcome these economic difficulties, it announced the 1986 New Economic Mechanism, which introduces capitalist elements; thus laying the groundwork for opening up the economy including privatization and attracting foreign investment. Especially since the collapse of the Soviet Union in the early 1990s, economic relations with capitalist countries have expanded further, transforming much of the economy into a form that depends on the aid of capitalist Western countries. Its inclusion in the global economy made it the 158th member of the WTO in 2013.

Despite these changes, Lao PDR still maintains its status as the world's poorest developing country. Lao PDR is a small country with population of 6.8 million, maintaining an agricultural-centered economic structure; primary products, including agricultural products, still account for a high proportion in the national economy but have very low productivity.

[Table 2-1] GDP and Economic Growth in Lao PDR

Indicator	Unit	2017	2018	2019	2020
Nominal GDP	1,000 Mil. Dollars	17.0	18.4	20.2	22.0
GDP/per capita	Dollar	2,555	2,720	2,931	3,156
Real Growth	%	6.8	6.5	6.7	6.8
PPP	%	0.6	2.0	3.1	3.3

Resources: IMF

Although the economy is small in size, it has continued to grow at an annual average rate of 6% over the past five years, and it is considered to have a solid growth engine (hydroelectric power generation, mineral) despite having no manufacturing base. Lao PDR has continued its fast economic growth rate with 6 percent, thanks to increased mineral and hydroelectric production, promotion of infrastructure construction projects, and booming tourism industry. Economic growth slowed slightly in 2018 compared to a year earlier due to massive flood damage but is expected to continue at a rapid rate of 6.7 to 6.8 percent between 2019 and 2020 based on increased power exports following the completion of hydroelectric power plants and large projects such as the China-Lao PDR high-speed railway.

1.2 National Development Strategy of Lao PDR²

The government of Lao PDR has set as its basic goal of economic management graduating from the "Least Developed Country (LDC)" by 2024 and growing into a "Middle-Income Country" by 2030. These national development strategies are embodied in the mid- to long-term vision, including vision 2030 and 10-year national implementation strategies for socio-economic development as well as the five-year national socio-economic development plan.

² OECD (2014); JST (2014)

1.2.1 National Socio-economic Development Plan (NSEDP)

In January 2016, the 10th Lao People's Revolutionary Party Convention (hosted every five years) elected a new leadership with a five-year term, including the first generation of the revolution by Bunnyang Borachit, secretary of the party, and adopted major documents such as the 8th five-year plan for national socio-economic development (2016-2020).

NSEDP 2016-2020 is based on the National Vision 2030 and the “10-year Socio-Economic Development Strategy,” which fall under the mid- to long-term strategy. Within the 8th NSEDP, the Laotian government states that, to graduate from LDC status, it will pursue policies to achieve three macro-performance objectives: continuous and comprehensive economic development in vulnerable areas, development of human resources, and strengthening of capacity to manage natural disasters.

Since 2015, the UN Evaluation for Graduation from being the poorest countries has surveyed the least developed countries for GNI per capita, Human Capital Index, and Economic Vulnerabilities Index every three years, and they graduated from being the poorest countries after a six-year review period (UN Economic Analysis & Policy Division, Graduation from being the Least Developed Country). Although income requirements (GNI per capita US\$1,229 or higher) are met, with the 2018 assessment meeting the requirements of the Human Capital Index (53.1 or higher) for the first time, the economic vulnerability index still does not meet the criteria (UN Economic Analysis & Policy Division, LDC at a glance).

The 8th NSEDP was established with focus on improving the UN's assessment indicators, especially with the aim of advancing the resource-based economic structure to boost economic operability and improve the human capital index. To this end, NSEDP presents priority tasks such as fostering industrial personnel through the development of special economic zones, enhancing efficiency in the financial industry, strengthening resilience against natural disasters, and utilizing sustainable natural resources. Regarding science and technology, the goals include fostering industrial personnel in the fields of construction, automobile repair, electronic engineering, urban engineering, mining, and geology, training science and technology personnel for industrialization, and improving researchers under

the Ministry of Science and Technology.

1.2.2 Vision 2030 and 10-year Socio-economic Development Strategy

As Lao PDR's long-term development goal, Vision 2030 aims to become a high-income country by 2030 through industrialization, human resource development, and sustainable use of natural resources.

Vision 2030 emphasizes the establishment of basic infrastructure for industrialization and modernization, development of human resources through the expansion of education and health services, and establishment of governance to utilize sustainable natural resources. The 10-year socio-economic development strategy, as the implementation strategy of Vision 2030, sets the goal of graduating from LDC status between 2016 and 2025, emphasizes industrialization, modernization, human development, and efficient utilization of natural resources as the strategy of Vision 2030, and suggests mid-term strategies such as improving governance and regional and international integration.

2. Lao PDR's National STI System

2.1 STI Policy Governance³

The Ministry of Science and Technology, which oversees science policy, was first launched in 1967 as the Ministry of Energy and Mining Industry; with the establishment of the National Authority for Science and Technology (NAST) in 1982, science and technology-related policies began to be discussed independently at the central ministry level. NAST was moved to the Prime Minister's Office in 2007 and promoted to the current Ministry of Science and Technology in 2011.

STI in Lao PDR is in the initial stage of governance establishment. The Ministry of Science and Technology is the government-centered one, and the relevant ministries, including R&D activities, Ministry of Agriculture and Forestry, and Ministry of Health are involved in the process of science and technology policymaking with focus on the research institute under its wing and the National University of Lao PDR.

R&D-related institutions were established in 1999 by the Ministry of Agriculture and Forestry (Ministry of Agriculture and Forestry), in 1995 by the National University of Lao PDR (11 departments), and in 2000 by the Institute of Health Science (7 research centers). The Lao PDR Pasteur Institute under the Ministry of Health is also operated as a major R&D institution.

2.1.1 Ministry of Science and Technology (MOST)

a) Function and role

The Ministry of Science and Technology (MOST) drafted the Prime Minister's Bill on Technology Transfer (2012), and it is responsible for drafting Article 24 of the Lao Constitution (2013) and National Science Week (2014), National Strategy Plan 2013-2020,

³ OECD (2014)

and Vision 2030 (2016-August) to play a central role in establishing and implementing national science and technology strategies (UNESCO (2018) GO->SPIN Country profiles in Science, Technology, and Innovation Policy).

The Lao government has been striving to apply science and technology to the overall social economy and state affairs with the aim of achieving SDGs through the vision of science and technology development 2020. Thus, the core task of MOST is to use science and technology to solve poverty problems and for Lao PDR to graduate from being one of the underdeveloped countries (UNESCO (2018) GO->SPIN Country profiles in Science, Technology, and Innovation Policy).

b) Major Organization

Through the reorganization in 2011, MOST was equipped with eight departments, four research institutes (ecology and plant diversity, renewable energy new materials, computers and electronics, technology management), two offices (National Science Council, vice minister's office), and 18 regional science and technology bureau systems.

1) Department of Technology and Innovation (DTI):

The Division of Technology and Innovation (DTI) was established under the Prime Minister's Decree on Organization and Activities (No. 0309/PM) of the Ministry of Science and Technology on September 28, 2011, with two directors, five departments (policy and strategy, administrative, innovation promotion, technology management, technology and innovation statistics) and two centers (high technology and innovation centers). To promote the utilization of a wide range of national technologies and innovations for production and business and service enhancement, national policies, strategies, legislation implementation, and achievement on technology evaluation, technical statistical data, development and promotion of innovation, human resources development, services, etc. are set as major business directions. The Korea Institute of Science and Technology Policy (2019) was referred to for the National Master Plan for Research and Development and Application of Science and Technology Innovation in Lao PDR.

The Department of Technology and Innovation (“DTI”) is under the Ministry of Science and Technology, established pursuant to the Prime Minister’s decree on organization and Activities of the Ministry of Science and Technology No. 309/PM dated 28 September 2011. The organization chart of DTI consists of a Director and two deputies of department, 5 Divisions, and 2 centers (Policy and strategy division; Administration division; Innovation promotion division; Technology management Division; Technology and innovation statistics division; Technology transfer promotion center; High technology center) with a total of 44 staff. The vision of the organization is to use Technology and Innovation nationally and widely to promote production, business, and service.

Its mission is to formulate and implement national policies, strategies, and legislation, carry out technology assessment, collect technology statistical data, and promote and develop innovation, human resource development, and service for enhancing national social-economic development effectively. The functions and duties are those of the secretariat for the Ministry on the development and macro management of policies and strategies in the promotion of technology and innovation. It was set up in the middle of year 2012 with the following functions and duties:

- Study national policies and strategies on technology and Innovation
- Formulate legislation on the management of technology and innovation
- Promote research on technology and innovation
- Provide services on technology and innovation
- Act as the central coordinating body with line ministries at the central and local governments on the management and monitoring of technology and innovation activities in the entire country
- Promote the transfer, application, and creation of technology and innovation

2) Department of Planning and Cooperation

The Department of Planning and Cooperation (“DPC”) is also under MOST, established in 28 September 2011 and last revamped on 26 February 2018. Currently, DPC has one Director

General and three Deputy Director Generals and consists of 7 Divisions and 1 Center: Administrative Division, ASEAN Division, International Organization Division, Bilateral Division, Planning Division, Evaluation Division, Domestic Cooperation Division, and Statistics Center. More than 40 staff are employed in DPC, whose vision is Service in plan, service in cooperation, and service in statistics.

Mission: Draft and evaluate the Ministry's plan to take part in the National Socio-economic Development Plan, collect and analyze all STI statistics, and act as the first door of the Ministry for cooperation in both domestic and international levels.

Function:

Coordinate directly with the Ministry of Planning and Investment on the MOST's report, plan on technical and infrastructure development using government budget and ODA in the STI sector.

- Coordinate directly with the Ministry of Foreign Affairs on the MOST's report and plan for bilateral, multilateral, and international organization cooperation.
- Collect and share STI statistics data with the Lao Statistics Bureau.

Promote cooperation between MOST and other partners such as bilateral, multilateral, and international organizations, enterprise, social organization, etc.

3) Cabinet of Academy of Science and Technology (CAST)

The Cabinet of Academy of Science and Technology ("CAST") is under the Ministry of Science and Technology, established under the Prime Minister's decree on organization and Activities of the Ministry of Science and Technology No. 309/PM dated 28 September 2011 and No. 0283/MoST on February 26, 2018.

The organization chart of CAST consists of a Cabinet Director and two deputy directors, 4 Divisions, and 1 center—Administrative and Service Division, Division of Planning and Cooperation, Division of Research Policy, Division of Information and Research, and

Research Center - Natural Science Research—with a total of 49 staff. As for the vision, mission, goal, and functions and duties, its vision is to promote STI research to improve the quality & quantity of the sector's production and services.

Its mission is to formulate and implement national policies, strategies, and legislation and carry out the resolutions of the ministries regarding technology, collect science and technology research statistics, promote STI research transfer to social use, human resource development, monitoring and evaluation of STI research projects, certify priority research areas related to sector strategy & research services provided for enhancing national social-economic development effectively. In the year 2025, CAST will move to an effective STI research organization in Lao PDR to improve the quality of life of the people as a follow-up to the national goal of graduating from being the poorest country to becoming a middle-income country.

Function and Duties:

CAST has functions and duties as the secretariat for the Science and Technology Ministry, National Science Council on coordination with the Sectoral Scientific Committee of Ministries and their Equivalents and Establishment by the Ministry of Science and Technology and research on presenting related proposals that consider and certify scientific and technological research results across the country. It was set up with the following functions and duties:

- Study, research and develop strategies, plans of the Party and the Government, plans and programs of the National Science Council (NSC) to become a project, planning, supervision, guidance, monitoring in implementation, research and development of laws, policies, regulations, rules, other guidelines for managing science research and technology development.
- Serve as the secretariat to the NSC in encouraging the establishment of the Sectoral Scientific Committee of Ministries, Equity Agencies, and Institutions.

- Research and propose to the NSC the approval and selection of scientific research, technological developments in accordance with the National Socio-Economic Development Plan at each stage of the Ministries and their Equivalents and Establishments, to compile the proposal of a research project on the sectors that are priority areas as well as to use scientific and technological research results that cause damage to society and economy, natural resources, environment, and practice of law-abiding citizens.
- Adviser promotion and strengthening to the science and technology research units for students and public and private scientists to participate in scientific and technological research, transfer of scientific and technological innovation, and promote and enhance the local wisdom in improving the quality of products and improving the quality of life of the people.
- Monitoring, evaluation, scientific and technological research of ministries, agencies, and organizations and collection of such reports.
- Train, transfer, create awareness, encourage research for socio-economic development.
- To coordinate with research, relevant units and summarize to the NSC for approval and certification of research, promote application achievements to the social community for use in socio-economic development.
- To cooperate, meet, and coordinate with the Scientific Conference of Asia, the region, and the world, and the organization for the development of Science and Technology.

In the past, the CAST never got ODA funding; in the future, we will apply for funding from the Republic of Korea.

2.2 Laws and Institutions related to STI

2.2.1 Process for Establishing STI Policies

Lao PDR's science and technology and R&D policies are established through five stages: agenda setting, policymaking, policy review, policy implementation, and policy evaluation.

In setting the agenda, the Ministry of Planning and Investment will prioritize national development through NSEDP, including agenda and STI activities. At this time, the STI portion of NSEDP goes through a process wherein MOST makes proposals and Congress reviews and approves them. Subsequently, the process of policy formation was carried out mainly by forming TFs among various ministries. At this time, the Ministry of Agriculture and Forestry, Ministry of Education and Sports, Ministry of Commerce, Ministry of Energy and Mining, and Ministry of Post are involved.

Once the policy draft is completed, the policy document will be reviewed at various stages before it is published and across all areas of government ministries, with the Prime Minister's Office announcing the policy or the law after parliamentary approval. The implementation of policies can be seen as research and innovation wherein individual activities consistent with the policies are carried out under the jurisdiction of the relevant ministries. The monitoring and evaluation of policies will then be conducted by the Planning and Investment Department, the Council, and the Ministry of Science and Technology together to assess policy inconsistency and achievement.

As shown in [Figure 4-3-2], various related ministries are participating in the policymaking process, indicating the importance of the role of MOST and the Ministry of Planning and Investment in the entire process. In addition, due to the nature of the political system, it can be seen that the National Assembly plays a major role in the process of setting agendas, making decisions, and evaluating policies.

2.2.2 Laws Related to Science and Technology Policies

The 2013 Science and Technology Act lays the basis of Lao PDR's science and technology policy. The law consists of a total of 69 clauses, specifying policies for promoting science and technology, budget, and role of MOST. The Science and Technology Act stipulates that science and technology policies should be established and implemented in accordance with the socio-economic development plan, government budget goals are defined, and MOST will play a key role in promoting science and technology.

2.3 R&D

2.3.1 R&D Expenditures

Lao PDR's total R&D spending is currently nonexistent, with the latest data at \$2.7 million in 2002 or 0.04 percent of GDP. At \$0.5, per capita R&D spending is very low similar to Cambodia.

In order to expand such R&D spending, Article 24 of the 2011 Science and Technology Act stipulates that the national science and technology budget must be 1 percent of the national budget; despite the fact that it is trying to increase investment in science and technology, only 1 percent of the national budget is allocated at a time when GDP itself is low.

Meanwhile, the Lao government research and development are carried out through the Science and Technology Development Fund (STDF). Established in 2014 to promote and support research and development, fund raising, innovation and technology transfer, and promote scientific and technological services, STDF supports researchers, young scientists, and students who carry out research and development activities that will contribute to solving the country's social, economic, and environmental challenges.

2.3.2 R&D Human Resources

Lao PDR's higher education can be divided into three years of junior college (undergraduate), four years of bachelor's degree (bachelor, four years), two years of master's degree (BA+2 years), and three years of doctor's (MA+3 years). In an effort to diversify the higher education system, the Laotian government has embodied the qualification of higher education, the bachelor's and master's courses, by the Ministry of Education's directives.

Based on this, the National University of Lao PDR (2019) was founded by the Asian Development Bank (ADB) in 1996 to integrate three existing universities in Vientiane (University of Education, Engineering, and Medical College) and nine higher schools (Electrical, Electronic, Electric, Transport, Irrigation, Forestry, and Agriculture) under the University Reform Plan of Asia Development Bank.

It has been making efforts to nurture higher education personnel. Despite these efforts, the total number of university students as of 2015 was about 770,000 according to the UNESCO (2018) Global Education Monitoring Report 2-17/18.

As of 2014, the ratio of PhD holders to the population remained very low at 0.01%. The number of Lao PDR science and technology researchers per 1 million people is about 76, and 19 of them are full-time workers, which is very low similar to Cambodia and Myanmar. The reason for the small number of researchers is that engineering, architecture, and IT are popular, but other science subjects are unpopular. Moreover, most people who have received college education believe that the fields of business administration and economics can be more beneficial than science (without the government's official scholarship program and incentive means for obtaining doctoral degrees in science and technology).

To overcome these limitations, the 8th NSEDP aggressively presented a goal of training 11 researchers per 10,000 people by 2020, or 14 times the number of people in 2012. Based on these plans and the results of an intergovernmental meeting with Vietnam in 2012, the government set up the 2018 Scientific and Technical Human Resources Training Center under the Lao Ministry of Science and Technology with \$44 million in support from Vietnam.

2.3.3 R&D Performance: Patents and Papers

In Lao PDR, 18 patents were filed in 2009, and 59, in 2018. The number of patent registrations began to be tallied in 2013, with one in the same year and five in 2018.

[Table 2-2] Patents in Lao PDR

Category		2014	2015	2016	2017	2018
Application	Domestic	1	1	-	3	1
	Foreigner	43	61	63	97	58
Total		44	62	63	100	59
Registration	Domestic	-	-	-	-	-
	Foreigner	-	2	2	3	5
Total		0	2	2	3	5

Resources: WIPO IP Statistics Data Center

Patent applications during 2001-2018 were mainly on medical technology, transportation, audiovisual technology, and pharmaceuticals (UNESCO (2018) GO-SSPIN Country profiles in Science, Technology, and Innovation Policy). Lao PDR's patent performance is very low as shown by the zero domestic patent registrations and 1.4 applications between 2014 and 2018.

2.3.4 ICT Environment

Lao PDR has the lowest digital environment among ASEAN countries in terms of Internet service accessibility, quality, and price. Compared to neighboring countries, Lao PDR ranks 139th among 179 countries in 2017 in the ITU's Informatization Development Index. Nonetheless, related industries and markets have expanded rapidly in line with the government's recent policy drive to develop the ICT sector.

[Table 2-3] Internet Access between City and Province (2017)

(Unit: %)

Category	City	Provincial
Mobile Access Rate per household	100	95
Fixed line telephone utilization	1.5	0.90
3G usage rate of Mobile	85	71
Wired Internet utilization	1.7	1.50

3. Lao PDR's Human Resources and STI Status

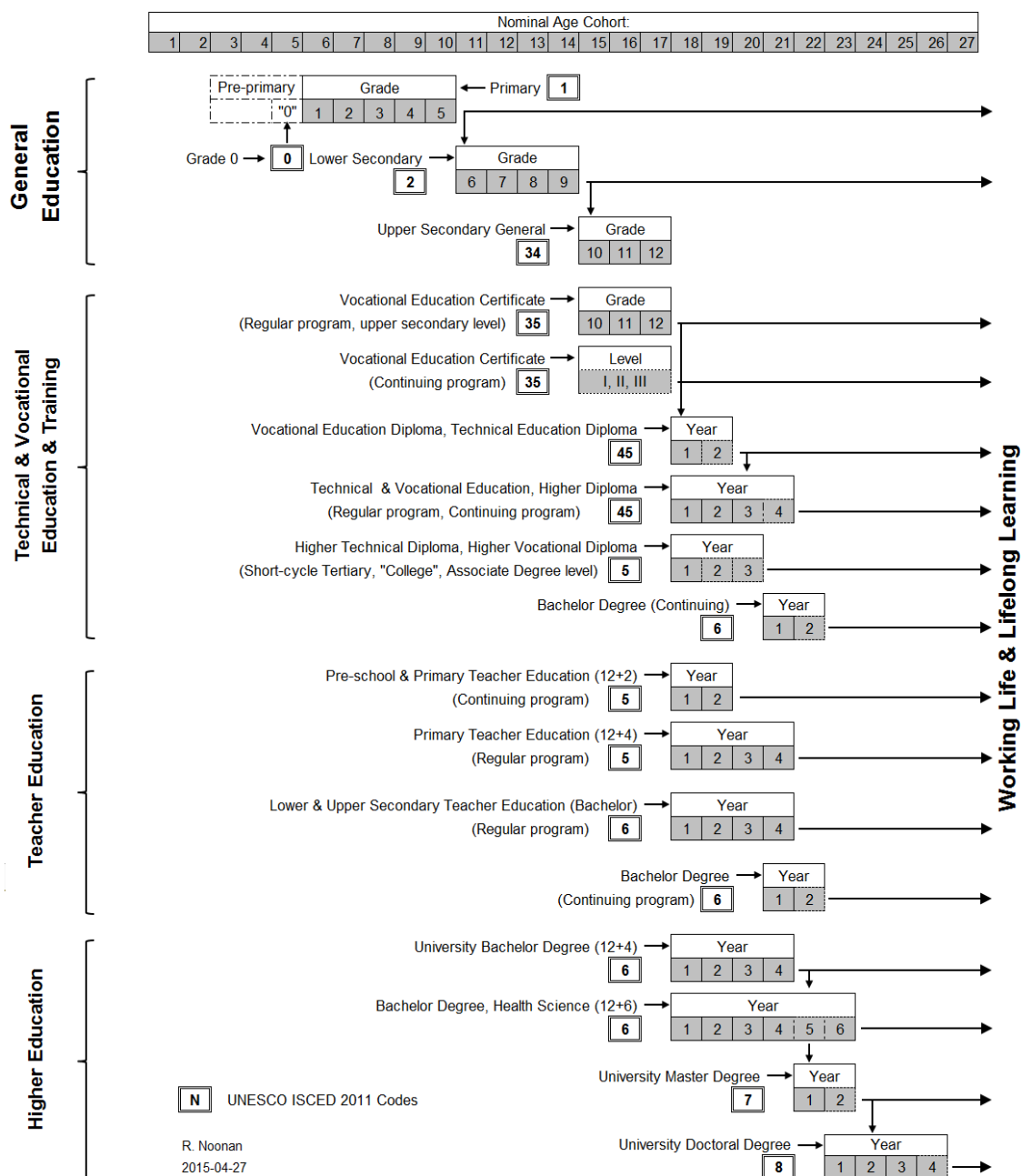
3.1 Education

3.1.1 Education System

a) Characteristics of Lao PDR's education system

- Lao PDR has five-year (elementary school), four-year (elementary school), three-year (high school), and four-year (college) courses for infant education, general education (elementary school), vocational and professional education, and higher education.
- Infant education includes child care education (three months to three years old) and kindergarten education (four to six years old), general primary education (one to five years old), lower secondary education (middle school course), and ten to twelve (high school course). Higher education has professional history (undergraduate, three years) and bachelor's degree (bachelor, four years).
- TVET: Technical Vocational Education & Training starts at the high school level, with the three-year high school courses consisting of Levels 1, 2, and 3, respectively, followed by the Diploma course as a junior college. Two-year courses are diploma, three-year courses are associate degree, and four-year courses are high school diploma.
- Universities for teacher training include two-year courses for infant and primary school teachers and four-year courses for primary school teachers, four-year courses for middle and high school teachers (four years, bachelor's degree), two years for master's degree and three years for PhD.
- From September to June of the following year, the school schedule is 33 weeks a year: 16 weeks for the first semester and 17 weeks for the second semester.

[Figure 2-1] Education System in Lao PDR



b) Status of primary and secondary education

- As of 2019, the total number of students in general education based on public schools was 1,547,622 students, 78,191 teachers, 12,424 schools, and 55,879 classes.

[Table 2-4] Lao PDR's General Education Status

Category	2018-2019 (Public)				2018-2019 (Private)			
	Class	School	Student	Faculty	Class	School	Student	Faculty
Kindergarten	5,657	2,923	7,739	191,261	1,732	509	3,801	39,875
Primary	32,074	8,570	33,533	717,853	1,918	284	2,358	52,806
Lower Secondary	12,308	897	25,041	426,822	591	64	1,186	16,034
Upper Secondary	5,840	34	11,878	211,686	197	-	395	5,308
	55,879	12,424	78,191	1,547,622	4,438	857	7,740	114,023

- As for the registration rate of general education, early childhood education has a low enrollment rate—only about 59 percent of child care centers and 79.4 percent of kindergartens. The number of elementary schools is close to 100%, but it is decreasing to 82.8% for middle schools and 54.8% for high schools.

[Table 2-5] Student Registration Rate in Lao PDR's General Education

Enrollment Rate	2018-2019		
	Total	Female	Male
Enrolment rate of 3 ~ 5-year-old children (%)	59,0	59,1	58,8
Enrollment rate of 5-year-old children (%)	79,4	79,5	79,4
Net Intake Rate in Primary Education (%)	98,2	97,9	98,4
Net Enrollment Rate in Primary Education (%)	99,1	98,8	99,3
Gross Intake Rate in Primary Education (%)	109,2	107,8	110,4
Gross Enrollment Rate of Primary Education (%)	108,2	107,0	109,4
Gross Enrollment Rate in Lower Secondary Education (%)	82,8	81,2	84,3
Gross Enrollment Rate in Upper Secondary Education (%)	54,8	52,0	57,5
Gross Enrollment Rate in Secondary Education (%)	70,9	68,8	73,0

c) Trend of college admissions

- Lao PDR has four universities under the Ministry of Education and one under the Ministry of Health; five universities are under the Ministry of Education, and there are universities in the field of education and universities in the field of technology, most of which are state-run, and private universities with the nature of professional universities.
- The number of college admissions for 2018-19 was 12,258, and the total number of college students was 39,497 (11,674 in 2017-18 and 36,447 students). The distribution of incoming students by major is as follows: 1) Education and Sports (1,126), 2) Language and Humanities (1,050), 3) Social Sciences (768), 4) Management and Law (Business, Management, and Laws (3,570), IT (118), and Science (Information and Communication, 118. There are 74 graduates in information and communication, 480 in science, 1,890 in engineering, 985 in agriculture, etc.

d) Major issues in 2020

- COVID-19 also affected Lao PDR's education. According to UNICEF data, more than 1 million students are not properly educated due to COVID-19. To this end, the Ministry of Education and Sports in Lao PDR has prepared countermeasures for COVID-19 related education with the support of UNICEF (KOTRA, 2020).
 - We are pushing for the establishment of a remote education system using satellite TV for teachers and students and the guarantee of continuous learning conditions through online platforms. Recently, we have been preparing in earnest through the "Master Plan for Digital Economic Development."
 - The online platform has an integrated system that links online courses, applications, TV, radio, YouTube, etc. according to the curriculum and provides TV and satellite receivers to provincial areas to prepare alternatives to avoid the spread of COVID-19.
- Lao PDR has formed a national resources committee headed by the deputy prime minister for education and has prepared a five-year development strategy (2016-

2020). The 8th National Assembly's 7th regular session (2019.6) is preparing a mid-term human resource development plan (2021-2025).

- The National Human Resources Development Committee is responsible for the training of socialist ideology for party leadership, the Ministry of Interior is in charge of training civil servants and administration and budget, the Ministry of Education and Sports is in charge of education, and the Ministry of Information, Culture, and Tourism is in charge of job creation and living values.

3.1.2 STEM Education

■ Status

- Basic Learning Competency (BLS) is taught in Lao PDR, where science is taught in the first grade in the subject "World Around Us." This course consists of science, social science (geography and history), environment, and health. It consists of three chapters: Living Things, Non-Living Things, and Lao PDR. Living Things and Non-Living Things include science, environment, and health, each consisting of the following details:

Category	Details
Living Things	1) Our Body and Our Health 2) Plants and their Uses 3) Animals and Their Behaviors 4) Relationship between Living Things
Non-Living Things	1) Matter and Its Uses 2) Energy and Chemicals
Lao PDR	1) Geography and History 2) Social Organization

- The areas of science are plants, animals, energy, and chemistry; especially for animals students learn classifications such as mammals and birds in the second grade as well as the characteristics of vertebrates and invertebrates in the fifth grade.

■ Overview of gifted education

- Lao PDR's education for gifted children (Gifted Education): The provisions concerning support for learners provide support for poor families and vulnerable groups (such as women and minorities), including support for students with outstanding results and good performance (Article 36 of the Lao PDR Education Act).
- Schools for gifted children are middle- and high school-level, with a changed boarding school for minorities established under the National Lao PDR University and a total boarding school established in the Southern Champasak area.
- Santhiphap School in Luang Prabang has opened a gifted class for the northern part of the country; a gifted class is operated in Vientiane, the capital city, for the central part of the country.
- There are also gifted classes in each province, mainly in Laotian language, Laotian literature, mathematics, physics, and chemistry. Some schools operate gifted classes in the same subjects.

3.1.3 TVET: Technical Vocational Education & Training

■ Overview

- The organizer of TVET in Lao PDR is the Ministry of Education & Sports, and each department has its own educational institution. The Ministry of Education, Sports, and Tourism has implemented vocational skills education and training based on the TVET Act, which was enacted in 2014, and the Ministry of Labor has implemented industrial technology standards and training policies for skilled human resources based on the Labor Act, which was amended in 2013.
- The Lao government is making efforts for the country to graduate from being one of the underdeveloped countries by securing skilled labor, and TVET plays an important role in this regard.

- Still, it is difficult to secure a competitive edge due to the lack of quality labor force in the corporate sector, which is also an important variable in securing Lao PDR's competitiveness in the process of integrating the ASEAN Economic Community (AEC).
- Currently, efforts are being made to reform Lao PDR's TVET system and train quality manpower, especially for this purpose, and Korea is steadily receiving support from Germany while providing support for the Ministry of Labor's vocational training projects.
- TVETs are provided to students who have completed elementary and middle school, with focus on producing middle- or high-ranking managers when students who have completed secondary school are admitted.
- There are three types of institutions (basic, intermediate, and advanced courses), and vocational education has skilled labor courses and functional qualification courses. Proficiency labor courses include three years after graduating from elementary school and three years after graduating from middle school. After graduating from high school, there are two-year courses and three-year courses.
- Vocational technology schools have 8+3 courses and 11+2 courses, and higher education institutions operate 11+3 courses and 11+5 courses for certificates or bachelor's degrees. After completing secondary education, graduates generally have five choices. They may enter the labor market, participate in technical training programs, participate in teacher training programs, enter universities or technical colleges, or go abroad to study.

■ Agency status

- As of 2019, the number of public and private educational institutions increased by 31,835; of these, 17,829 were under the Ministry of Education, and the rest were trained by institutions of related ministries. A total of 6,458 students were admitted to courses in agriculture, finance, health, information, culture, and tourism, with 7,548 admitted to private training institutions (a 10% increase from the previous year).

- There are 75,466 students, 14,153 graduates from public and private educational institutions in 2018, 1,513 graduates from agriculture, 6,212 students from industry, and 6,428 students from services.
- Under the Ministry of Education, there are 90 public and private educational institutions with 52,278 (women 26,255), and 4,526 (women 1,615) students, respectively. There are 23 public schools with 35,611 (women 14,519) and 2,294 (women 854) students, and 67 private schools with 21,667 (women 11,736) and 2,232 (women 761) students. Majors in private institutions were mainly management, foreign language, and IT.

[Table 2-6] Total Number of Public TVET Students under MoES

No.	Sectors	Total Number of Students	Percentage
1	Agriculture	3,047	8.56%
2	Industry	19,709	52.35%
3	Service	12,544	35.23%
4	TVET teachers	311	0.87%
Total		35,611	100%

[Table 2-7] Total Number of Private TVET Students under MoES

No.	Sectors	Total Number of Students	Percentage
1	Agriculture	0	0%
2	Industry	2,921	13.48%
3	Service	18,746	86.52%
Total		21,667	100%

3.2 Human Resource Development Strategy

■ Overview of the national development strategy

- Lao PDR has established the “National HRD to 2025” to establish a five-year human resource development strategy every year, and it is in the process of reviewing the performance of HRDS (2016-2020) and establishing the next strategy (2021-2025) in 2020.
 - The vice ministers of 28 ministries and agencies serve as members of the committee including the deputy prime minister and education minister. There is a secretariat in the Ministry of Education, which is responsible for the division of planning and investment.
- Lao PDR has limitations in terms of finance, manpower, culture, technology, etc. in human resource development. According to the HRD, which has been promoted for 10 years since 2001, securing financial support through cooperation with donor countries is a priority for national human resources development because of the insufficient financial support for education and health services and lack of support for civil servants. Human resources have low level of education and knowledge, especially limited ability to speak English. Especially in the mid-70s, many intellectuals went abroad, so there was a shortage of leadership staff. Therefore, it is necessary to train manpower to overcome this situation.
- It is not easy to secure educational opportunities in the rural areas due to the cultural environment where ethnic minorities are large, and there are some environments wherein women do not have many opportunities. The growing consumption of alcoholic beverages, large and small car and motorcycle accidents, and some ethnic groups preferring to work at home rather than go to school are also obstacles to development.
- The technological aspect, especially the development of satellite broadcasting and the Internet, is also a constraint on human resource development. For example, various forms of broadcasting in Thailand are being transmitted to the home turf, affecting education, culture, and society as a whole.
- Human resource development is very important as a cornerstone of national development after overcoming these limitations.

■ Main contents of the Lao PDR HRD Strategy (2016-2020)

- Mission

The purpose of the HRD Strategy is to complement sectoral strategies and plans by focusing on (i) problems that require multi-sectoral solutions and (ii) cross-cutting issues that affect the implementation of both sectoral and multi-sectoral plans and strategies.

- Core principles

“Man is not only a resource for socio-economic development but also the core element of development itself” (HRD Strategy to 2020). “Education is the starting point and core of human resource development” (HRD Strategy to 2020). All people should have equal opportunity to develop their skills according to their abilities. People who are unable to secure their basic needs through work and other family resources should receive assistance from their communities and the government.

- Strategic priorities

The main objective of the HRD Strategy to 2025 is to provide broad social sector support to Lao PDR’s overall development goal “of continued poverty reduction, graduation from Least Developed Country Status through realization of national development potential and comparative advantages, effective management and utilization of natural resources, and strong regional and international integration” (8th NSEDP).

- Goal

The goal of the HRD Strategy to 2025 is: (i) graduation from Least Developed Country Status (the core objective of the 8th NSEDP) and (ii) achievement of off-track HDR-related Millennium Development Goals (MDG).

Graduating from least developed country status involves three indicators (corresponding to the three Outcomes in the 8th NSEDP): (i) level of gross national income (GNI) per capita; (ii) index of HRD outcomes called the Human Activity Index (HAI); and (iii) index of risk to exogenous shocks called the Economic Vulnerability Index (EVI). The HRD-related HAI (the

key performance indicator for Outcome 2 in the 8th NSEDP) is an equally weighted index of four HRD indicators: (i) percentage of the undernourished population (based on FAO estimates); (ii) under-5 mortality rate; (iii) gross secondary school enrollment rate; and (iv) literacy rate among adults aged 15 and above (higher values are needed for graduation). Although the first two indicators have been improving steadily over time, the two education indicators have been improving more slowly due to the low primary school completion rate. At the last review (2012), Lao PDR's HAI had a value of only 61.4 compared to the graduation threshold value of 66 at the time.

The off-track HRD-related MDGs include: (i) percentage of children under 5 who are moderately short for their age; (ii) gross secondary education enrollment ratio; (iii) proportion of pupils starting grade 1 who reach the last grade of primary school; (iv) literacy rate among 15~24-year-olds; (v) infant mortality rate; (vi) proportion of 1-year-old children immunized against measles; and (vii) maternal mortality ratio.

- Strategic framework

The draft strategic framework has six strategic objectives, each of which includes two or more strategic thrusts. Although some of the strategic objectives appear to be associated most closely with a particular sector (e.g., education, labor, health), they all address multi-sectoral problems that require multi-sectoral solutions.

- ① Reduce and eventually eliminate disparities in HRD outcomes
- ② Support HRD through key multi-sectoral investments in education and training
- ③ Increase labor productivity
- ④ Support HRD through key multi-sectoral investments in health and nutrition
- ⑤ Preserve and enhance traditional culture, promote tourism, and foster sports development
- ⑥ Protect the population from catastrophic risks

In addition to the six strategic objectives, the draft strategic framework includes four cross-cutting themes for investments that are needed to address important constraints to achieving HRD objectives in multiple sectors:

- ① Political leadership
- ② Civil service capacity
- ③ Gender equality
- ④ Regional cooperation

• Monitoring Plan

Goal						
	Human Asset Index (HAI)	61.4	2012	68*		UNCTAD
	Children under 5 who are moderately short for their age (%)	44.2%	2011/12	28.0%		Lao Social Indicators Survey (LSIS)
	Gross secondary school enrollment ratio (%)	45.8%	2011	80.0%		Lao Social Indicators Survey (LSIS)
	Pupils starting grade 1 who reach the last grade of primary school (%)	70.0%	2012	90.0%		EMIS, Ministry of Education (Lao PDR)
	Literacy rate of youth aged 15-24 (%)	73.1%	2011/12	99.0%		Lao Social Indicators Survey (LSIS)
	Infant mortality rate	68	2009/10	30		Lao Social Indicators Survey (LSIS)
	Children aged 12-23 months immunized against measles (%)	55.0%	2011/12	95.0%		Lao Social Indicators Survey (LSIS)
	Maternal mortality ratio	357	2009	200		Lao Social Indicators Survey (LSIS)
SO-1						
	Under-5 child mortality rate (PAR)	63.7%	2011/12	51.0%	40.8%	Lao Social Indicators Survey (LSIS)
	Children under 5 who are moderately short for their age (PAR)	55.4%	2011/12	44.3%	35.5%	Lao Social Indicators Survey (LSIS)
	Percentage of children aged 36-59 months who are currently enrolled in preschool (PAR)	217%	2011/12	150%	100%	Lao Social Indicators Survey (LSIS)
	Literacy rate, youth ages 15-24 (PAR)	30.8%	2011/12	24.6%	19.7%	Lao Social Indicators Survey (LSIS)

Goal						
SO-2						
	Government-financed expenditure on education as a percentage of total primary government expenditure	17.1%	2013/14 (planned)	19.0%	19.0%	Ministry of Finance <i>Gazettes</i>
	Non-salary recurrent expenditure on education as a percentage of total government-financed expenditure on education	10.9%	2013/14 (planned)	25.0%	25.0%	Ministry of Finance <i>Gazettes</i>
	Children aged 3-5 who are currently enrolled in pre-school (%)	23.0%	2011/12	55.0%*		Lao Social Indicators Survey (LSIS)
	Literacy rate, youth ages 15-24 (%)	73.1%	2011/12	99.0%*		Lao Social Indicators Survey (LSIS)
	Employers citing "inadequately educated workforce" as their main constraint (%)	17.0%	2012	15.3%	13.8%	World Bank Enterprise Survey
SO-3						
	Labor productivity (i.e., GDP per worker) in manufacturing (million Kip, 2002 prices)	16.1	2013	18.0	20.0	National Accounts (GDP), LECS (employment)
	Number of registered foreign workers in Lao PDR	20,942	2011/12	25,130	30,156	Ministry of Labor (Lao PDR)
SO-4						
	Government-financed expenditure on health as a percentage of total primary government expenditure	6.9%	2013/14 (planned)	9.0%	9.0%	Ministry of Finance <i>Gazettes</i>
	Non-salary recurrent expenditure on health as a percentage of total government-financed expenditure on education	45.9%	2013/14 (planned)	45.0%	45.0%	Ministry of Finance <i>Gazettes</i>
	Those who are moderately short for their age among children under 5 (%)	44.2%a	2011/12	28.0%*		Lao Social Indicators Survey (LSIS)

2020 K-Innovation ODA Program with Lao PDR

Goal						
	Male adults (aged 15+) who smoke tobacco (%)	48.0%	2011	42.0%	36.0%	WHO
	Average annual number of liters of pure alcohol consumed per adult aged 15 and above	7.3	2008-2010	6.0	5.0	WHO
	Annual number of motor vehicle accident deaths	910	2013	850	800	Ministry of Public Security (Lao PDR)
	Number of deaths from major disease epidemics in Lao PDR during the previous 5 years	77	2010-2014	0	0	EM-DAT. The International Disaster Database (www.emdat.be)
SO-5						
	Annual number of tourist arrivals in Lao PDR (millions)	4.15	2014	6.0		Ministry of Information, Culture, and Tourism (Lao PDR)
	Annual income from tourism (US\$ millions)	641	2014	953		Ministry of Information, Culture, and Tourism (Lao PDR)
	Cultural villages as a percentage of all villages	39.0%	2015	65.0%*		Ministry of Information, Culture, and Tourism (Lao PDR)
	Percentage of women aged 15-49 with comprehensive knowledge of HIV	23.0%	2011/12	28.8%	35.9%	2011-12 Lao Social Indicators Survey (LSIS)
	Percentage of the population exercising regularly or participating in sports			30.0%*		Ministry of Education and Sports
	Number of medals won by Lao PDR in the most recent SEA Games	79	2013	87	96	Ministry of Education and Sports (Lao PDR)
SO-6						
	Average percentage of the population affected by natural disasters during the previous 5 years	3.7%	2009-2013	3.0%	2.5%	EM-DAT database (www.emdat.be)

Goal						
	Population with health insurance (%)	18.10%	2012	80.0%*	100.0%	Ministry of Health (Lao PDR)
	Out-of-pocket expenditure on healthcare as a percentage of total healthcare expenditure	46.0%	2009/10	40.0%	20.0%	WHO (National Health Accounts for Lao PDR)
CCT-1						
	Corruption Perceptions Index (worldwide ranking)	140	2012	130	120	Transparency International
	Ease of Doing Business Index (worldwide ranking)	159	2013	150	140	World Bank
	Number of female members of the Politburo	1	2015	2	2	Central Party Bureau
CCT-2						
	Expenditure on government salaries and compensation (Chapters 10+11) as a percentage of total government revenue	50.5%	2013/14 (planned)	30.0%	30.0%	Ministry of Finance (Lao PDR) <i>Gazettes</i>
	Female share of civil servants who are Director Generals, Deputy Director Generals or equivalent	16.70%	2012	20.0%	25.0%	Ministry of Home Affairs (Lao PDR)
	Civil servants with bachelor's degree of higher (%)	26.70%	2012/13	32%	40%	Ministry of Home Affairs (Lao PDR)
	Average civil service salary as a percentage of GDP per capita	170%	2014			Ministry of Home Affairs (Lao PDR)
CCT-3						
	Female share of paid employment (%)	34.6%	2010	37.0%	40.0%	Lao Labor Force Survey
	Female share of National Assembly members (%)	25.0%	2012	32.5%	40.0%	National Assembly (Lao PDR)
	Female share of Ministers, Deputy Ministers, Assistants	10.7%	2012	15.0%	20.0%	Central Party Bureau

Goal						
	to Ministers or equivalent (%)					
	Female share of Provincial Governors or Vice District Governors (%)	6.8%	2012	10.0%	15.0%	Central Party Bureau
	Female share of District Governors or Vice District Governors (%)	6.1%	2012	10.0%	15.0%	Central Party Bureau
	Female share of Village Chiefs or Deputy Village Chiefs (%)	4.8%	2012	7.0%	10.0%	Central Party Bureau
CCT-4						
	Percentage of Lao students studying abroad who are studying in an ASEAN or a GMS country	82.1%	2012	84.0%	86.0%	UNESCO (www.uis.unesco.org)
	Percentage of foreign skilled workers in Lao PDR who are from GMS countries	86.0%	2011/12	86.0%	86.0%	Ministry of Labor (Lao PDR)

- Implementation Plan (2016~2020)

SO #1. Reduce and eventually eliminate disparities in HRD outcomes			
ST 1.1. Identify and invest in gifted children from the disadvantaged population groups at an early age	1. Ministry of Education and Sports The 8th NSEDP (2016-20) includes several programs targeting disadvantaged children (especially those in disadvantaged districts), including school block grants, building of student dormitories, school lunch and food supplements, and scholarships to facilitate access to higher education for students from provinces that do not have a university and for students from poor families.	2016-2020	NA 8 th NSEDP
	2. SURAFCO Project This project promotes the enrollment of youth and ethnic people in the National Agriculture and Forestry College (NAFC) by providing scholarships to young women and to non-Lao-Tai ethnic people. It also supports the development of curricula that are more practical and relevant in the northern uplands environment.	2009-2016	CHF 12 million (Phases I and II combined) (Swiss Agency for Development and Cooperation)
	3. National high schools This project will develop a system of national high schools for girls and boys of exceptional abilities in provinces where there is no university in order to prepare them to enter the university and excel in their studies. All students will be admitted based on merit only, receiving full scholarships, and students residing more than 10	2016-2020	NA

	kilometers from the school will receive free room and board. There are no plans at present to develop national high schools, but model secondary schools are being constructed in each province and will be linked with an ICT Center at MoES.		
ST 1.2. Educate all children and youth to become effective parents	1. Life Skills Curriculum for Rural Lower Secondary School Students This project will develop and pilot a life skills curriculum for rural lower secondary students to prepare them for adolescence and eventual parenthood. The curriculum will focus on life skills, including reproductive health, preventive health (e.g., avoidance of drugs, alcohol, and tobacco, importance of exercise), and early childhood development and learning (e.g., preventive healthcare, nutrition, support for learning, access to learning materials).	2016-2020	NA
ST 1.3. Strengthen the demand for HRD investments among disadvantaged groups	1. School Feeding Program The main objective of the School Feeding Program is to provide incentives for school enrollment and attendance, especially in remote areas. Nonetheless, the program has multiple objectives, including improving children's health and nutrition (ST 4.1), and involves multi-sectoral cooperation in its implementation. The World Food Program (WFP) is planning to support the School Feeding Program through 2018 when it will begin a gradual handover of the program to the government.	2016-2020	\$9 million per year (WFP through 2018)

2. Ministry of Education and Sports	2016-2020	NA 8th NSEDP
The 8th NSEDP includes several initiatives (in addition to the school feeding program) designed to strengthen the demand for HRD investments among disadvantaged groups, including: (i) per capita block grants to schools to enable informal fees to be abolished; (ii) scholarships targeting disadvantaged children in grades 6-9 (beginning with disadvantaged districts); and (iii) targeted (“pro-poor”) scholarships for 20% of upper secondary, TVET, and university students.		
3. Ministry of Health	2016-2020	NA 8th NSEDP
Continue support for free maternal and child health (MCH) services for all women and children under 5 years and not currently covered by other social health protection schemes; implement measures to increase the demand for MCH and reproductive health services (e.g., incentives, vouchers, conditional cash transfers, incentives for timely referrals, maternity waiting homes, reimbursement of travel costs, and food costs of inpatients).		
4. Street Children Project	2015-2020	\$60,000 per year (Aide et Action)
This project, with technical support from Aide et Action, aims to re-integrate street children into their families and into the public school system by offering income-generating activities to poor parents.		
5. Immunization services for remote rural areas	2016-2020	NA

	This project will support the piloting of regularly scheduled immunization camps in remote rural villages where immunization rates are currently very low. The immunizations will be provided by a mobile team based in a nearby town, and the visits will be announced ahead of time to villagers by village health volunteers. A small incentive (e.g., 1 Kg bag of rice) will be provided to parents for each immunization, with an additional small gift after the child is fully immunized.		
SO #2. Support HRD through key multi-sectoral investments in education and training			
ST 2.1. Get children off to the right start through effective early childhood education (ECE) targeting disadvantaged rural children	1. Ministry of Education and Sports Develop basic infrastructure for extending kindergartens to villages without permanent schools. Improve the quality of preschool teacher training. Continue block grants to preschools.	2016-2020	NA 8th NSEDP
	2. Early Childhood Education Project This project will support ECE activities, including multi-sector community-based components focusing on health (including disabilities), nutrition, and information dissemination.	2015-2019	\$28 million (World Bank)
	3. Early Childhood Development and Education This project, with technical support from Aide et Action, supports selected schools in ECE, promotes nutrition (school lunches) and hygiene, develops and applies child development monitoring tools and	2016-2020	NA (Aide et Action)

	mobilizes parents to promote and practice positive disciplines, child rights, and protection and participate in ECE activities.		
ST 2.2. Build job-related skills that employers want	1. Second Higher Education Strengthening Project (MoES) As the Lao component of a GMS initiative to strengthen universities in Cambodia and Lao PDR, this Project will focus on issues of higher education (HE) quality and relevance, including: (i) quality and relevance of academic programs; (ii) research capacity; (iii) equitable access to HE (including support for a new campus at Savannakhet University, scholarships for female and disadvantaged students, and pilot student loan program); and (iv) governance and management (including a review of the regulatory and funding framework for private HE institutions). The project design will be supported by an ADB project-preparatory TA project in 2015.	2015-2020	\$40 million (Lao component) (Asian Development Bank)
	2. Second Technical and Vocational Education Project As the Lao component of a GMS initiative to strengthen TVET in Cambodia, PRC, Lao PDR, and Myanmar, this project is expected to strengthen formal and non-formal TVET institutions, adopt competency-based training modules, provide incentives to attract more students to appropriate fields of study, and promote public-private partnerships in skills development. The Project will support the development of curricula	2016-2020	\$25 million (Lao component) (Asian Development Bank)

	and TVET teacher training consistent with regional standards. The project design will be supported by a project-preparatory TA project in 2015.		
	3. Lao-Japan Human Resource Development Institute (LJI) Supported by the Japan International Cooperation Agency (JICA) and based in the National University of Lao PDR (NUOL), this project will establish LJI as a focal point for the development of business skills in support of the private sector, with focus on small and medium-sized businesses in the vicinities of Vientiane Capital and Savannakhet Province.	2014-2019	NA
	4. VELA Project This project aims to increase TVET enrollments in 11 provinces, targeting 2,000 apprenticeship places in companies and 10,000 youth from rural areas. The project includes two mechanisms to strengthen TVET enrollments: an “Inclusive Access Fund” covering both study costs for students and course costs for schools; and a “Social Marketing” initiative to change parents’ and students’ attitudes toward manual work. The project will also (i) focus on developing close cooperation between employers and the TVET sector, (ii) strengthen the capacity of the National Training Council, (iii) develop professional standards, and (iv) provide incentives to employers to create new in-company training places.	2014-2017	EUR 2.9 million (Swiss Agency for Development and Cooperation)

	5. Support for Hospitality Skills Development SDC supports two programs to develop hospitality skills for the tourism sector. The first program provides support to LANITH to allow youth to acquire the skills needed to work in hotels and restaurants. The second program is expected to provide co-financing of a Luxdev project, with SDC supporting the low end of the hospitality skills development spectrum (apprenticeships, short courses, and in-service training toward certificates that can eventually lead to diplomas).	2015-2020	CHF 2 million per year (LANITH program only) NA (co-financing of Luxdev project)
	6. Initiative for Livelihood Education and Development (ILEAD) This project, which is based in Vientiane Capital and supported by Aide et Action, provides training to youth ages 15-35 on market-oriented trades/skills, soft skills, and entrepreneurial skills so that they can access better livelihood opportunities.	2015-2010	\$80,000 or more per year (Aide et Action)
SO #3. Increase labor productivity			
ST 3.1. Improve the performance of labor markets	1. Ministry of Labor and Social Welfare Strengthen recruitment services for new workers; develop skills standards for 25 professional areas, test and issue certificates of skill standards, and develop the labor market information system.	2016-2020	NA 8th NSEDP
ST 3.2. Develop evidence-based migration policies	1. Ministry of Labor and Social Welfare	2016-20	NA 8th NSEDP

	Develop appropriate policies on (i) minimum wage, (ii) use of foreign workers in areas that are short of domestic workers, (iii) labor movement within and outside the country, and (iv) shifting workers from agriculture to the industry and service sectors.		
SO #4. Support HRD through key multi-sectoral investments in health and nutrition			
ST 4.1. Reduce the prevalence of stunting among young children	1. Ministry of Health In its nutrition project, MOH will procure and distribute Vitamin A, iron folate, and deworming pills to women and children and promote breastfeeding and appropriate maternal and child nutrition across the country. It will also monitor children's growth and provide local nutritious food supplements to malnourished children. In addition, the project will strengthen the leadership skills of the National Nutrition Committee and increase involvement and coordination in multiple sectors and mass organizations.	2016-2020	NA 8th Health Sector Development Plan
	2. Mother and Child Health and Nutrition This program is designed to reduce the prevalence of stunting among vulnerable children. The program is currently supported by the World Food Program (WFP), providing nutritional assistance to 22,000 pregnant and nursing mothers and children under 2 out of 16 districts of 3 provinces.	2016 (but may continue during 2017-2020 if funding is available)	\$1 million per year (WFP)

ST 4.2. Delay the onset of non-communicable diseases	1. Ministry of Health Increase health promotion (particularly among youth, women, and low-income individuals) on the dangers of smoking and inhaling smoke and risks from excessive consumption of alcohol, importance of avoiding salty, fatty, and sugar-laden food, and importance of regular exercise. Continue to strengthen the control of tobacco and alcohol consumption. Collaborate with MOES to develop curricula on the dangers of tobacco and alcohol consumption at the primary and secondary levels. MOH will also develop a multi-sectoral strategy on NCD prevention and control.	2016-2020	NA 8th Health Sector Development Plan
ST 4.3. Ensure public health security	1. Greater Mekong Subregion (GMS) Health Security Project The project will focus on the prevention of communicable diseases, including HIV/AIDS, tuberculosis, food-borne diseases, and neglected tropical diseases (e.g., dengue, schistosomiasis, lymphatic filariasis, blinding trachoma, and helminthiasis). Activities are expected to focus on the following areas: (i) communicable disease surveillance; (ii) diagnosis and treatment of communicable diseases; (iii) strategies for addressing the special risks of mobile and migrant populations; and (iv) regional coordination, knowledge sharing, and capacity building for communicable disease control.	2017-2021	\$13.5 million (Lao component) (Asian Development Bank)

	2. GMS Capacity Building for the HIV/AIDS Prevention Project	2013-2017	\$5 million (Lao component) (Asian Development Bank)
SO #5. Preserve and enhance traditional culture, promote tourism, and foster sports development			
ST 5.1. Develop cultural heritage sites to promote tourism	1. Greater Mekong Subregion Tourism Infrastructure for Inclusive Growth Project The project will be implemented in four provinces and is expected to increase tourism employment and revenue in those projects. The project will focus on paving rural roads to open new areas for tourism, improve access to markets and social services, improve environment conditions in important tourism areas, and support the expansion of small and medium-sized tourism businesses.	2015-2020	\$43.57 million (Asian Development Bank \$40 million, Government \$3.57 million)
ST 5.2. Strengthen mass media to support HRD	There is nothing in the 8th NSEDP on the information/communications sector. Check with the MICT focal point to see if there is a national communication strategy. Strong mass media support is necessary especially for most of the other Strategic Thrusts (most notably for 1.3, 3.1, 4.1, and 4.2).	2016-2020	NA
ST 5.3. Develop broad-based participation in sports	1. Ministry of Education and Sports Develop the capacity of athletes and gymnasts to meet regional and international standards, encourage all people in society to participate in sports and gymnastics (especially youth, women, and elderly); and promote and develop indigenous sports.	2016-2020	NA 8th NSEDP

SO #6. Protect the population from catastrophic risks			
ST 6.1. Disaster prevention and mitigation	1. Ministry of Natural Resources and Environment Complete the establishment and improvement of network stations and early warning systems in river basins. Map areas that are particularly vulnerable to the risk of floods, droughts, and other natural disasters. Enhance self-initiatives toward disaster risk management. Increase the ability to adapt to and mitigate the effects of climate change. Improve legislation on the use and management of natural resources. Participate in regional and global efforts to prevent and mitigate the effects of natural disasters.	2016-2020	NA 8th NSEDP
	2. Project on Climate Change (Ministry of Health) This project will establish a storm warning system and a disease precaution system and conduct climate change impact assessments. It will support the capacity building of central and provincial staff on the management and control of health emergencies arising from natural disasters. The project will also reduce and respond to the spread of diseases as a result of natural disasters, provide emergency water and sanitation equipment to local levels, and prepare for any disaster-related nutrition emergencies.	2016-2020	NA 8th Health Sector Development Plan
ST 6.2. Protect the population from	1. National Health Insurance Organization (MOH)	2016-2020	NA (external support)

astronomical healthcare costs	The government has a long-term objective of insuring the entire population against the risk of ill health and astronomical healthcare expenditure through the National Health Insurance (NHI) scheme, which combines the four previously existing social health insurance schemes. The focus of NHI work during the period 2016-2020 will be on (i) developing, implementing, and piloting the newly established NHI scheme and (ii) expanding coverage.		expected from EC, WHO, and SDC)
CCT #1. Political leadership	Lao People's Revolutionary Party Disseminate policy directions, laws, and regulations and educate, women, youth, juveniles, and ethnic people on political thinking for their broad and deep understanding and active contribution to the country's development. Implement the Three Builds Policy. Promote strict adherence to laws and regulations to deter social problems such as corruption, burglary, human trafficking-prostitution, and drug use/trade. Improve and expand the Lao People's Revolutionary Youth Union to be able to lead, educate, and consolidate the youth, juveniles, and ethnic youth with the guidance of the Party.	2016-2020	NA 8th NSEDP
CCT #2. Civil service capacity	1. Ministry of Home Affairs Upgrade government officials in all sectors on politics and governance, technical skills, and foreign languages, including female staff, ethnic staff, and staff who are underprivileged. Improve and further develop the	2016-2020	NA 8th NSEDP

	training curriculum for government officials to be more systematic and to meet international standards. Promote a good, healthy environment in government workplaces. Make greater use of information technology in personnel management. Improve personnel manuals and strengthen the review of staff performance. Update each period policies on staff compensation to ensure their consistency with the actual cost of living and living conditions. Implement a policy on staff promotion and salary increases linked to capacity and performance. Implement a strategy for gender equality and provide opportunities for women and ethnic groups as well as the underprivileged to join government service at both central and local levels.		
CCT #3. Gender equality	1. 8th Five-Year Plan for the Advancement of Women Train trainers to strengthen capacity of NCAW at all levels as well as the secretariats focusing on NCAW at the provincial, district, and village levels to eliminate all forms of discrimination against women. Develop families, villages, and districts committed to gender equality.	2016-20	NA 8th NSEDP
	2. Action Plan for the Prevention and Elimination of Violence against Women and Children Advocate and raise awareness throughout society and strengthen the capacity of officials to prevent and address violence against women.	2014-2020	NA 8th NSEDP

	Develop families, villages, and districts committed to domestic nonviolence especially against women and children.		
CCT #4. Regional cooperation	1. Implementing the Greater Mekong Subregion (GMS) Human Resource Development Strategic Framework and Action Plan (SFAP Phase 2) This regional technical assistance project will support the following activities in Lao PDR and several other GMS countries: (i) further development of a regional framework for the mutual recognition of skills and qualifications in three areas (i.e., welding, automotive technology, housekeeping); (ii) development of a system for the mutual recognition of training standards for TVET teachers and trainers in the same three additional skill areas; (iii) development of standard learning materials for TVET; (iv) facilitating subregional cooperation in establishing quality assurance systems in TVET; (v) development of an Academic Credit Transfer System in GMS universities; (vi) promotion of GMS university networking by supporting the startup of the GMS University Consortium for universities that are not members of the existing ASEAN University Network; (vii) support for the establishment of a GMS knowledge platform, (viii) enhancing social protection for migrant workers in the GMS; (ix) monitoring the implementation of the MOU Action Plan to Reduce HIV Vulnerability Related to Population Movement in the GMS; (x)	2014-2017	\$1,75 million (ADB \$1.25 million, PRC \$0.50 million)

	strengthening GMS cooperation in anti-human trafficking; (xi) assessing the social impact of increased GMS connectivity; and (xii) support for the country-level implementation of the GMS SFAP Phase 2, including the development of national HRD action plans, monitoring, and annual reporting to the GMS Working Group on HRD.		
	2. GMS HRD Strategy and Action Plan (next phase)	2018-2020	NA

3.3 Educational Environment of Science, Technology, and Engineering

a) Higher Education Institution

■ Status

- In the five-year development plan for the development of higher education in Lao PDR, 51 strategies are established under 7 goals: 1) education linked to the market and socio-economic needs; 2) promotion of opportunities for higher education and lifelong education; 3) provision of social services; 4) improvement of higher education institutions to international levels; 5) strengthening the research activities of research institutes; 6) improvement of planning and operation; and 7) strengthening of international cooperation.
- Higher education is divided into undergraduate (undergraduate, 3 years), bachelor's (bachelor, 4 years), master's (BA+2 years), and PhD (MA+3 years).
- Lao PDR has 103 institutions of higher education, 5 national universities, 4 academies, 3 research institutes, 20 colleges, 71 private universities, and 11 educational colleges.

- Lao PDR's first national university was the National University of Lao PDR (NUOL) in Vientiane, the capital city, founded in 1996 with the integration of 10 institutions of higher education and supported by the Asian Development Bank.
- There were 8,137 enrolled students including 2,500 enrolled students in the only PhD program in Lao PDR at the time of its establishment.
- Since then, universities have been established in the southern, northern, and central regions to foster local talent. Champasak University was founded in 2002 in Paxe, the southern state of Champasak, Suphanouvong University in the northern city of Luang Prabang in 2003, and Savannakhet University in Central Savannakhet in 2010.
- Separately, the University of Health & Science was established in 2007 under the Ministry of Health and Human Services, independent of the National University of Lao PDR.
- The 4 universities have 3,053 employees (1,788 women) and teachers and 33,109 students (1,151 students). There are two types of college admissions: one is free of charge, with about 80% of the total selected based on local and regional recommendations, and the other 20% attending by paying tuition fees.
- There are 270 courses in 4 universities, 3 of which are PhDs and 42 are master's degrees. The Higher Education Act was enacted in accordance with the Education Act in 2008, the development plan was established every five years, and the strategies and policies for scientific research and research operation were established and operated in 2015.

■ **Limits of higher education in Lao PDR**

- The Laotian higher education is deemed to be lacking efficient and responsible management of public educational institutions, basic data on higher education, and coordination between public and private sectors.
- Moreover, given the demand in the labor market, there are constraints on implementing long-term strategies for higher education; in particular, national educational institutions need curriculum and learning materials to promote the National Education System Reform Strategy (2006-2015).
- Despite the need for university infrastructure and hands-on equipment, especially libraries, research laboratories, and equipment, the reality is dismal. Supanobong University in the northern part of the country has infrastructure with the support of the Korean government, but Champasak University in the southern part of the country has few such infrastructure. As such, ADB is providing support to Champasak University and Sabanak University in accordance with the “High Education Competency Enhancement Project.”
- In addition, the policy department of the Ministry of Education became independent in 2008, and its capabilities and bases are very weak. Moreover, support for the autonomy and finance of each university is not being provided properly, and credit recognition, quality control, and research and development are urgently needed. In the case of Lao PDR National University, a University Steering Committee is formed, but Supanobong University does not have such functions; hence the difficulty in operating the school. Nonetheless, a Korean expert-centered advisory committee has been formed to provide advice.
- The quality of the university is determined by the teaching and research capabilities of the instructor (professor), and the degree is represented in this capacity. Because the current ratio of chair and doctorate holders to Laotian university professors is very low, however, the university's educational research capabilities are low, especially in the case of local national universities where bachelor's degree instructors are giving lectures. Accordingly, the Ministry of Education, Sports, and Tourism has set a goal of improving

the ratio of university professors to “1 (doctor):6 (doctoral):3 (doctoral)” by 2015 in order to improve the quality of higher education, but the project progress has been slow due to lack of budget.

■ National University of Lao PDR

- As the first comprehensive university in Lao PDR, National Lao PDR University has been carrying out its education and learning base with the support of the ADB on November 5, 1996. In 2019, it has been established as the largest comprehensive university in Lao PDR with 23,015 students (2018-19), 1,808 teachers, and 346 employees.
- NUOL consists of 5 campuses, 13 colleges, 2 research institutes (Institute of Lao-Japan Human Resource Development and Institute of Chinese Confucius), 1 central library, and 6 centers. The five campuses are named after their respective regional names, including the Dongdok campus, the Sokpaluang campus, the Nabong campus, and the Donnokkhum campus.

■ Supanubong University

- Supanobong University was established in 2003 with the aim of expanding higher educational opportunities in eight Northern Lao PDR states (Luang Prabang, Sayabouly, Xiengkhouang, Phongsali, Luang Namtha, Bokeo Oudomxay, and Houaphan) and promoting the industry in the northern part of Lao PDR through human resource development.
- The campus was completed in November 2007 under the Korean government's Assistant Secretary of State (EDCF), with support such as software and hardware supply and expert dispatch, curriculum and teaching materials development, teaching staff training, book supply and consulting (subject departments: agriculture, forestry, livestock studies, tourism management, international management, IT management, computer engineering, electrical engineering, architectural engineering).
- The main purpose of the establishment of Supanobong University was to establish a university that was most suitable for the realities of Lao PDR and Luang Prabang to

combine the know-how of the Korean education system and to become a successful model of the university in terms of operation through software support such as university operation, curriculum and teaching materials, and teaching staff training, instead of simply supplying hardware such as equipment and facilities. Supanobong University is now developing into the second largest national university in Lao PDR, attended by 4,500 students.

- The organization structure of SU consists of 1 president and 2 vice presidents. The administrative office is composed of offices of general administration, finance, academic affairs, audit, organization and personnel, student affairs, construction and service, planning and international coordination, research and scientific services, and library and ICT.
- SU has 6 faculty members and 34 programs including 2 master's degree programs, about 3,500 students (34 students are pursuing master's degrees), and 300 lecturers. PhD holders number 10, 161 staff and lecturers have a master's degree, and others have a bachelor's degree or a high school diploma.
- The Faculty of Education offers 6 bachelor degree programs: Lao language and literacy, English, mathematics, physical, political science, and chemistry. The Faculty of Economics and Tourism has 8 programs: economics, international business management, general business management, IT business management, tourism management, hotel management, finance and banking, and accounting. The Faculty of Agriculture and Forest Resources offers 5 programs: plant science, animal science, forest resources, food science and technology, and eco-tourism. The Faculty of Engineering has 6 programs: electrical engineering, computer engineering, civil engineering, material engineering, logistic technology, and electronics and communication. The Faculty of Architecture offers 3 programs: architecture, interior design, and Lao architecture. The Faculty of Languages has 4 programs: Lao language, general English language, Korean language, and Vietnamese language. SU also has an institute and some centers such as Confucian institute, Korean language center, and Lao-Korean business incubating center.

No.	Faculty	Doctoral		Master		Bachelor		Diploma	
		T	F	T	F	T	F	T	F
1.	Education	1	0	29	4	39	18	0	0
2.	Agriculture and Forestry	3	0	38	8	17	7	0	0
3.	Economics and Tourism	2	1	34	11	23	6	0	0
4.	Architecture	1	0	14	2	18	6	0	0
5.	Engineering	2	0	27	2	20	3	1	0
6.	Languages	1	0	19	8	19	7	0	0
	Total	10	1	161	35	136	47	1	0

No.	SU	Doctoral		Master's		Bachelor's		Diploma	
		T	F	T	F	T	F	T	F
1.	Office of Academic Affairs	0	0	4	2	6	3	1	0
2.	Office of Administrative Affairs	0	0	7	1	5	2	4	0
3.	Office of Personnel Affairs	0	0	4	2	9	0	0	0
4.	Office of International Relations	0	0	9	3	2	1	0	0
5.	Office of Student Affairs	0	0	3	1	6	3	3	2
6.	Office of Research and Academic Service	1	0	5	2	3	2	0	0
7.	Office of Financial Affairs	0	0	1	0	5	2	4	3
8.	Office of Infrastructure and Campus Management	0	0	4	0	5	1	0	0
9.	Office of Inspection	0	0	2	0	3	1	0	0
10.	SU Library	0	0	6	2	4	2	1	1
11.	Confucius Institute	1	1	2	1	2	1	0	0
12.	LKST	1	1	2	0	0	0	0	0
13.	ICT	0	0	3	0	5	1	0	0
		3	2	52	14	55	19	13	6

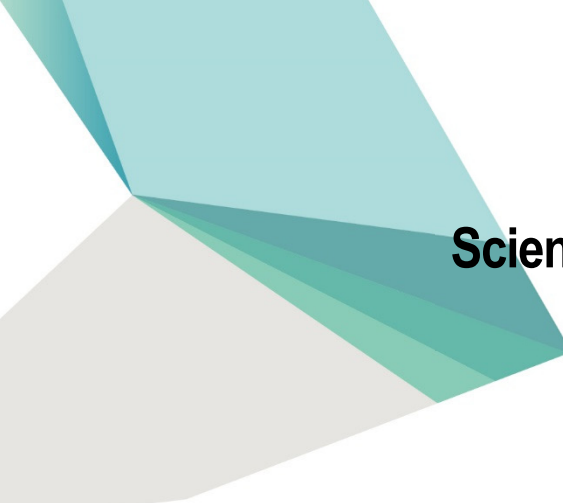
No.	Faculty	Enrollment					
		Bachelor		Master		Continuing Program	
		T	F	T	F	T	F
1.	Education	534	243	9	2	76	20
2.	Agriculture and Forestry	569	312	-	-	-	-
3.	Economics and Tourism	941	504	25	8	-	-
4.	Architecture	382	77	-	-	-	-
5.	Engineering	576	148	-	-	-	-
6.	Languages	295	159	-	-	-	-
	Total	3,297	1,443	34	10	76	20



CHAPTER 3

Sharing Korea's Experience in Developing Science and Technology Policies as a Tool for Economic Growth

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022



Chapter 3. Sharing Korea's Experiences in Developing Science and Technology Policies as a Tool for Economic Growth

1. Overview of Korea's Science and Technology Policy

1.1 Introduction

Korea's economic development has been described as "compressed growth." Formerly the poorest country in the world and a country with \$87 per capita income in the 1950s, Korea is now the 10th largest economy in terms of GDP and a member of the OECD (as of 2020). Remarkably, Korea has developed an indigenous R&D capability that has allowed the export of state-of-the-art goods, such as semiconductors, LCD panels, and smartphones. Among many factors that have contributed to the economic development of Korea, the policies associated with science and technology (S&T) are regarded as crucial (Yoon, 2014).

The Korean government has developed and adopted a number of powerful policy instruments to advance quickly and boost the level of S&T growth capability that ultimately led to economic development. The government has tried to foster S&T R&D in the private sector through a range of policy approaches, such as tax relief, incentives, loans, staff, and procurement. It has also set up numerous technology creation funds in the forms of direct subsidies, technology loans, and investment. In addition, the government has made attempts to resolve human resource problems by offering improved research and financial assistance using a specialist researcher framework. Korea's approach to technological acquisition has had both positive and negative results.

On the positive side, the strategy of Korea allowed it to procure technology at a lower cost

⁴ Ji Woong Yoon (2014), Evolution of science and technology policy in Korea, *The Korean Journal of Policy Studies*, Vol. 29, No. 1 (2014), pp. 147-172.

and overcame the restrictions frequently placed by multinationals on local companies wishing to develop their own skills. The strategy has been successful in preserving freedom from the domination of multinationals. On the negative side, Korea had to give up access to emerging technology that could have been made possible by direct ties with international companies. By restricting foreign direct investment, Korea lost the opportunity to follow global norms on the domestic market (Chung, 2010). The most important lesson here is that, had it not been for the well-trained workforce, Korea would not have been able to absorb and assimilate foreign technologies by technology import (Yoon, 2014).

The government has played significant role in the development of Korea's Science and Technology. As the main features of the Korean policy measure, first, the Korean government focused on the expansion of R&D investment and human resource of S&T from the very early stage of development, which is considered a key factor that contributed to the rapid growth of Korea's S&T. Second, the government put much effort to determine the policy direction and optimal growth path. Third, it focused on fostering the R&D player/subject: Government Research Institutes (GRI) in the 1970s; private companies in the 1980s, and universities in the 1990s. Moreover, it actively promoted the national R&D program providing support measures as well as infrastructures at the same time. Finally, the Korean government sought the productivity of input resource by implementing measures of selecting the strategic sector and concentrating investment on the available resources.

This chapter aims to describe the experience of Korea's development of Science and technology policy. As an overview of the long-term and mid-term S&T policy throughout the development of Korea, from the 1960s to the current administration (as of 2020), we will also examine Korea's experience of selecting strategic industries and sectoral procedure of the National Technology Roadmap (NTRM) and development of human resource in the S&T policy, which has been considered to be key for the compressed growth of the country.

The Master Plan for Science, Technology, and Innovation in Korea consists of a long-term plan (every 15~20 years), a mid-term plan (every 5 years, an inter-governmental plan), and a short-term plan (every year). Korea's first Science and Technology policy was implemented in 1962 (the 1st 5-year Technology Promotion Plan, 1962~1966) in line with Korea's 1st 5-

year Economic Growth Plan. The "Long-Term Plan for the 1980s" (1967~1986) was announced as the first long-term S&T plan.

The Science and Technology Policy Institute (STEPI) explains the evolution of the Science and Technology Policy in five stages: 1) Formation of S&T Foundation (1945~1966); 2) Establishment of comprehensive S&T administration and technology development system (1967~1970s); 3) Technology drive policy and expansion of the national R&D system (1980s); 4) Promotion of S&T strategies for advanced technology development (1990s); and 5) Strengthening the ability to lead science and technology through creation and innovation (2000s~). Meanwhile, Yoon (2014) roughly divided Korea's development stages of Science and Technology (hereafter S&T) policy into three stages: Set up (1960s~1970s), Learning to Developing (1980~1997), and Enhancing the Innovation System (After). Shin, et al (2012) analyze these in a similar way: Creation of platform (1960s~1970s), Expansion and Deepening (1980s~1990s), and Toward an Efficient Innovation System (2000s and after).

1.2 Long-term Science and Technology Policy

The Long-term Science and Technology Policy in Korea has played a pivotal role in setting the direction of Korean S&T. "The long-term plan of the 1980s" (1967~1986) was drafted based on the vision of reaching the level of mid-industrialized country by the middle of the 1980s. The "Long-term plan for 2000s" (1987~2001) set the vision of ranking among the world's TOP 10 technology advanced countries and executed the program of 5 technologies (information, materials, energy, resource, welfare), 2 institutions (human resource, support measures), and 6 strategies (R&D personnel, target technologies, R&D expenditures). The "2025 Vision" (2000~2025) set the vision to be the top seven in science and technology competitiveness by 2025. To this end, a step-by-step objective to create global technological leadership in selected fields by 2025 was proposed to ensure superior science and technology competitiveness over Asian rivals by 2005 including a research center in Asia Pacific by 2015. The "2040 Vision" (2011~2040), with a view to making Korea the world's fifth largest global science and technology leader, proposed that the development of potential core technologies in 25 fields and 5 policy positions be divided into 3 phases.

Recently, the Korean government launched the “S&T Future Strategy 2045” (Aug. 2020) with the future forecast of 1) security and health (biotech, medicine, environment), 2) convenience (AI, manufacture innovation, energy, mobility), 3) communication and trust (digital infrastructure), and 4) contribution (aerospace, etc.)

As such, STI has been at the core of Korean development from the very beginning. The First Five-Year Economic Plan (1962–66) was supplemented with the “Five-Year Science and Technology Development Plan,” which contained plans for skills training, technology acquisition, and S&T capacity building. Indeed, STI has been a key strategy for attaining the national development goals in the series of Five-Year Economic Plans, which placed S&T on top of the development agenda throughout the entire development period (1961–1991).

1.3 Mid-term Science and Technology Policy

1.3.1 Early Stage of Laying the Foundation (1960s~1970s)

In the 1960s and 1970s, the government aimed to establish an institutional foundation for the development of S&T. Accordingly, S&T policy was a crucial part of the five-year economic development plan, with the relevant laws and institutions created. In the latter part of the 1960s, the Ministry of Science and Technology (MST) and a government-funded research institute were established to implement policies for fostering the advancement of S&T. Moreover, the Korean government focused on learning about and absorbing foreign technologies by licensing and importing capital goods. According to Chung (2017), the Foreign direct investment (FDI) was not a preferred method of financing economic growth, as the public perceived FDI as being dependent on foreign countries’ technology and systems.

The Korean government proliferated technological learning through various policy measures in the 1960s–1980s/1990s. To begin with, it created demand for new technologies by consistently implementing industrial development projects that opened up new business opportunities for private enterprises. In order to seize opportunities, private

enterprises had to acquire the necessary technologies in a competitive manner. This was particularly so in the 1970s when the government pushed ahead with the development of heavy chemical industries, which were considered very technology-intensive by the Korean standard of the 1970s (Chung, 2017).

[Table 3-1] 5-year Science and Technology Development Plan During the 1960s~1980s⁵

Plan	Vision and Goals / R&D / Strategic Industries
1 st 5-year Technology Promotion Plan (1962~1966)	- Vision: Secure 601,763 personnel in technology to complete the “the 1st 5-year economic development plan,” lay the foundation to reach the level of an industrialized country - R&D Investment: KRW 520 million ('61) -> KRW 1,360 million ('66) *%/GDP: 0.25%-> 0.5%
2 nd 5-year Science and Technology Promotion Plan (1967~1971)	- Vision: 1) Maximizing manpower (human brain) for creativity and function development for productiveness, 2) Enhancing Science and Technology independence by promoting research activities, 3) Adopting Advanced Knowledge of S&T to promote industrial development and S&T capacity, 4) Creating a scientific climate - R&D Investment: KRW 41.46 billion - Strategic Industries: 11 industrial technologies (mining, metal, machine, shipbuilding, chemical, textile, electricity, nuclear, civil engineering, medicine and sanitation, agriculture and forestry) and 6 basic sciences (math, physics, chemistry, biology, astronomy, meteorology)
3 rd 5-year Science and Technology Promotion Plan (1972~1976)	- Vision: 1) Development of Heavy and Chemical industries, 2) Significant increase in exports, 3) Innovative development of the rural economy, 4) Guarantee of National Security, 5) Creation of scientific climate - R&D Investment: total of KRW 142.7 billion for 5 years - Strategic Industries: Heavy chemical industry (machinery, metal, electronic, chemical industry) and infrastructure, Implementation of large-scale national project for shipbuilding, marine resource, and nuclear (includes establishment of the relevant research institutes)

⁵ We referred to and translated Moon, *et al* (2011), Assessment of Korea's Science and Technology Mid- and Long-term Plan

Plan	Vision and Goals / R&D / Strategic Industries
4 th 5-year Science and Technology Promotion Plan (1977~1981)	- Vision: 1) Enhancing independent technology development capacity through quality improvement of S&T manpower and R&D development, 2) Promoting Economic Development through the strategic development of advanced industrial technologies to foster brain-intensive industries and technology innovation, 3) Creating and fostering an S&T climate through national dissemination
	- R&D Expenditure: KRW 79.6 billion ('77) → KRW 243.2 billion ('81) *GERD%/GDP: 0.7% → 1.5%, Public/Private: 70:30 → 50:50
	- Strategic Industries: Fine chemical, precision machinery, plant, knowledge-information technology

1.4 Policy Shift from Technology Learning to Developing (1980~1997)

In the second phase (1980~1997), under the administrations of CHUN, ROH, and KIM, "The Korean government shifted its policy direction from technology learning to the growth of its own science and technological capability in the hi-tech sectors that needed indigenous R&D capabilities" (Yoon, 2014). The government has initiated large-scale R&D initiatives aimed at a variety of fields. By benchmarking the policies of advanced economies, the government has been able to target promising S&T areas. Korea consequently made significant investments for taking advantage of economies of scale in some sectors, such as the semiconductor, electronics, steel, and chemical industries.

1.5 Enhancement of Systematic ST Innovation System (late 1990s~)

Korea has sought to synthesize and strengthen the structure of S&T technologies. In the late 1990s, the country experienced a financial crisis that transformed the country's socio-economic structure. There are a variety of anecdotal theories given as to what caused the crisis. One is that Korea's economic development was too rapid, and that the country did not have a sound, solid foundation to sustain economic growth. While the infrastructures involved have been in place for 40 years and the innovation capacity for R&D has grown, they have not been seen to work together effectively, which is essential for sustainable

innovation. The government responded to this problem by developing a policy aimed at synthesizing and systemizing the entire S&T innovation mechanism (Yoon, 2014⁶).

[Table 3-2] 5-year Science and Technology Development Plan During 1982~2002⁷

Plan	Vision and Goals / R&D / Strategic Industries
5 th 5-year Economic and Social Development Plan: S&T Sector Plan (1982~1986)	- <u>Vision</u>: Reaching the technology level of advanced countries in the 1980s through rapid growth of technology development - <u>R&D Expenditure</u>: KRW 713.3 billion ('82)→ KRW 1468.6 billion ('86) - * GDP ratio (%): 1.3%→ 2.0%, Public/Private: 49:51→ 45:55 - <u>Strategic Industries</u>: Semiconductors and Computers, Fine Chemical, Machine industry, other industry (biology, metal, textile, plant, etc.)
6 th 5-year Economic and Social Development Plan: S&T Plan (1987~1991)	- <u>Vision</u>: Rank among the world's TOP 10 technologically advanced countries. In specific sectors, reach the world's best level - Vision: Building a base and achieving a milestone for growth as an advanced technology country in the 2000s - <u>R&D Expenditure</u>: KRW 2,036.6 billion ('87)→ KRW 4,196.8 billion ('91) - * GDP Ratio (%): 2.2%→ 3.0% - <u>Strategic Industries</u>: Information Technology (Computer, Software, Semiconductors), Energy and resource, Fine chemical, Biotechnology, New materials development, Engineering, Environment, Marine, Aerospace
7 th 5-year Economic and Social Development Plan: S&T Sector Plan (1992~1996)	- <u>Vision</u>: Rank among the 7 advanced countries in S&T in 2000s in S&T resources (lower than 7 advanced countries), Productivity and Technology (mid-lower than 7 advanced countries), Basic technology and S&T culture (OECD level) - <u>14 Strategic Key Technologies</u>: Next-generation product technology development (Ultra-high Integrated Semiconductor, ISDN, HDTC, Electric Car, AI computer, New medicine, Hi-tech production system), Source-based technology (Information-Electricity-Energy, hi-tech mobility and parts, Biotechnology, Environmental engineering, New energy, Nuclear, etc.)

⁶ Ji Woong Yoon (2014), Evolution of science and technology policy in Korea, The Korean Journal of Policy Studies, Vol. 29, No. 1 (2014), pp. 147-172.

⁷ We referred to and translated Moon, *et al* (2011), Assessment of Korea's Science and Technology Mid- and Long-term Plan (Reference in Korean)

Plan	Vision and Goals / R&D / Strategic Industries
New Economy 5-year plan: Strategy for Technology Development (1993~1997)	- <u>Vision</u>: Build the solid foundation for advancing S&T to reach the level of 7 advanced countries by the early 21st century - <u>Strategic industries</u>: Follow-up and development of 5-year industry technology plan and 5-year plan for localization of machine, parts and material, Aerospace, Transport, and environmental technology
5-year plan for S&T (1997~2002)	- <u>Vision</u>: Promoting the total capacity of science and technology to reach G7 level in the early 21st century by developing original technology skill - <u>Strategic Industries</u>: IT, Robotics, Machinery, Material, Communication, Energy and resource, next-generation battery, environmental, medical equipment, brain research, transportation, Engineering, SOC and National Security-related technology
Revised 5-year plan for STI (2000~2002)	- <u>Vision</u>: Promoting the total capacity of science and technology to reach G7 level in the early 21st century, 1) reach world-class level as a future promising industry, 2) have R&D expenditure exceed 5% of GDP, 3) reach world's TOP 10 level of basic technology by 2002 - <u>Strategic Industry</u>: rearrangement of existing 6 industries to 3 categories (key tech, hi-tech, social welfare) 12 technologies

1.6 Establishment of Institutional Basis and First Comprehensive Government-wide Plan for S&T Development (2002~)

In the 1990s, Korea first implemented S&T foresight projects to generate strategic intelligence on science and technology. S&T foresight has rapidly spread through different industries. S&T foresight has enabled building strategic R&D projects on a wide scale. On the other hand, both private sector and government have witnessed a paradigm change in technological innovation from the IT sector (Shin, et al, 2012⁸). In January 1999, the Kim administration established the *National Science and Technology Committee (NSTC)* as the highest decision-making and coordinating body of Science and Technology policy. The president appointed himself as chairman for ensuring full political support for the Science

⁸ Taeyoung Shin, Sungju Hong, and Juyoung Kang (2012), Korea's Strategy for Development of STI Capacity: A Historical Perspective, STEPI, Policy Reference 2012-01.

and Technology policy coordination. Due to the number of government entities who were involved in planning and conducting R&D programs, and the amount of the national R&D funds had rapidly enlarged, it was important to keep cross-governmental policy coordination for the R&D programs to be aligned with the national policy agenda. In 2001, the NSTC proliferated a long-term plan for Science and Technology development and created a framework act to implement this plan. In addition, based on the 1988 Science and Technology law, a basic S&T plan was made, the first comprehensive government-wide plan for Science and Technology development. In turn, based on this plan, technology roadmaps for 99 selected technology fields were drawn up (Chung, 2010; Yoon, 2014).

The “Science and Technology Promotion Act” enacted in 1967 led to the creation of the Korea Science and Technology Foundation and promotion of technology advancement, but there were weaknesses in the promotion of the development strategy in response to new demands for science and technology in the coming era. In addition, the “Special Act for the Advancement of Science and Technology” introduced in 1997 was provisional until June 2002, thereby requiring alternative legislation (STEPI, 2017). Since the enactment of the Framework Act on Science and Technology (Jan. 2001), the government has been establishing a basic plan that comprehensively combines the science and technology-related plans and policies of the relevant central administrative agencies every five years in accordance with Article 7 of the Framework Act on Science and Technology. The plan is finalized after deliberation by the National Science and Technology Committee.

The five-year Science and Technology Plan had not been upgraded from the part of the Economic Development Plan to an independent legislative plan until the creation of the Act. Indeed, until the 1990s, the mid- and long-term science and technology plan was a supplementary factor to support the promotion of the economic growth plan (Moon, et al, 2011). If this was the essence of a supplementary human resources support plan, the Science and Technology Structure Act was established, and the Science and Technology Specific Plan was established based on this Act.

[Table 3-3] Article 7 of the Framework Act of S&T in Korea

Article 7 of the Framework Act of S&T clearly states that “The Government shall set mid- and long-term policy objectives and directions related to the development of science and technology in order to achieve the purpose of this Act efficiently and shall finalize them after deliberation by the National Science and Technology Council under Article 9 (1).”

Moreover, it requires the formulation of a master plan for science and technology every five years. According to Article 7 of the Framework Act of S&T, “(2) The Minister of Science and ICT shall formulate a master plan for science and technology (hereinafter referred to as “master plan”) every five years, considering the mid- and long-term policy objectives and directions under Paragraph (1) and plans, policies, etc. for science and technology formulated by the relevant central administrative agencies, and shall finalize the master plan after deliberation by the National Science and Technology Council in accordance with Article 9 (1).”

[Table 3-4] Science and Technology Basic Plan (2002~2022)

Plan	Vision and Goals
1 st S&T Basic Plan (2002~2006)	- <u>Vision</u>: Economic growth to the level of USD 15,000 per capita and harnessing the competitiveness to reach the world’s top 10 S&T level by 2006 - <u>R&D Expenditure</u>: KRW 35,000 billion for the next 5 years - <u>Strategic Industries</u>: 6 promising new technologies (IT, BT, NT, ST, ET, CT) and 20 core targets, 77 key technologies for development
S&T Basic Plan for Participative Government (2003~2007)	- <u>Revised</u> version of existing plan, adopting the NRTM - <u>Vision</u>: Enter the second phase of S&T-based country by S&T-oriented society / To rank among the World’s Top 8 Countries in the field of S&T - <u>R&D Expenditure</u>: KRW 4,268.9 billion (’01)→ KRW 35,31.6 billion (’03~’07)
2 nd S&T Basic Plan for Advanced Country (2008~2012)	- <u>Vision</u>: 577 initiative (expanding GERD up to 5%/GDP, support for 7 core industries, to rank among the world’s Top 7 S&T countries) - <u>Mission</u>: Economic growth and job creation, Creative Innovation System - <u>Strategic Technologies</u>: 50 critical and 40 candidate technologies

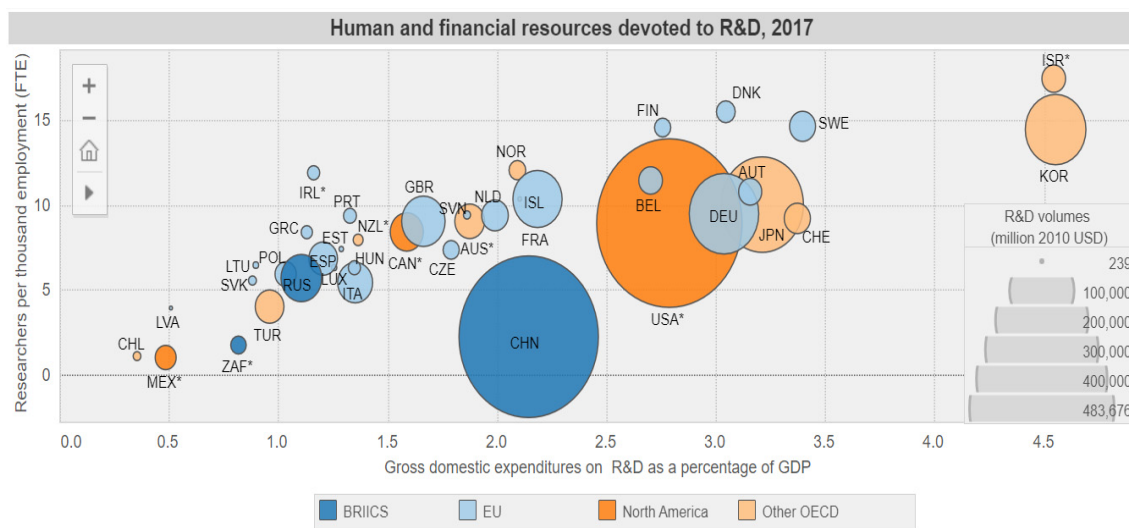
Plan	Vision and Goals
3 rd S&T Basic Plan (2013~2017)	- <u>Vision</u>: New era of hope with Creative S&T - <u>Goal</u>: Strengthening STI capability (become the world's TOP country), Economic growth and quality of life (complete 40% of R&D contribution of economic growth) as well as Job creation (64,000 jobs)
4 th S&T Basic Plan: S&T Innovation and Challenges for 2040 (2018~2022)	- <u>Vision</u>: Improving the quality of people's lives and contributing to the development of human society through science and technology - <u>4 strategies (19 tasks)</u>: 1) increase S&T capability; basic research of KRW 1.26 trillion ('17) to double by 2022, interest index for S&T from 37.7('16) to 45('22), 2) STI innovation ecosystem; innovative start-up (%) from 21%('14) to 30%('22), number of industry-academe-research joint patent/per 1000 researchers 2.3('14) to 3.0('22), 3) New Industry and job creation led by S&T; 260,000 jobs created based on S&T and ICT by 2022, 4) happy life by S&T; ratio of healthy elderly people 21.1% ('15) → 25% ('22), fine dust ratio in Seoul $26 \mu\text{g}/\text{m}^3$ ('17) → $18 \mu\text{g}/\text{m}^3$ ('22)

2. STI Ecosystem in Korea

2.1 Korea's Global Landscape for R&D

Despite the global economic slowdown, global spending in R&D has continued to rise consistently following the financial crisis of 2008 (Chung, 2017). According to OECD estimates, world R&D spending/Real R&D expenditure in the OECD region increased by 3.9% in 2018, mostly driven by business R&D results that accounted for more than 75% of this development. Accounting for 71% of the total R&D performance in the OECD region, the Business Enterprise market saw its R&D spending increase by 4.2% in 2018. Korea has steadily expanded its R&D spending over the last four decades. In 2018, Korea was one of the largest R&D performers in the OECD area along with the United States, Japan, Germany, and France (OECD, 2020).

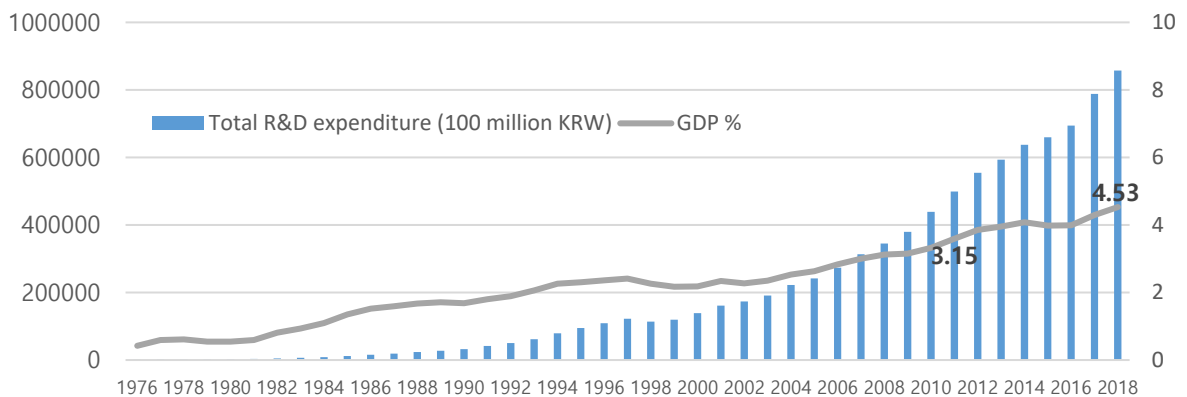
[Figure 3-1] Global R&D Structure



Source: OECD, Database of Main Science and Technology Indicators (www.oecd.org/sti/rds).

Korea's total R&D expenditure has dramatically increased from KRW 60.9 billion in 1976 to KRW 3.21 trillion in 1990 and KRW 13.85 trillion, eventually reaching KRW 85.73 trillion in 2018. The ratio for GDP(%) demonstrated significant growth, from 1.68%/GDP in 1990, 2.18%/GDP in 2000, and 3.32%/GDP in 2010 to 4.53%/GDP in 2018, which is the World's top tier among OECD countries.

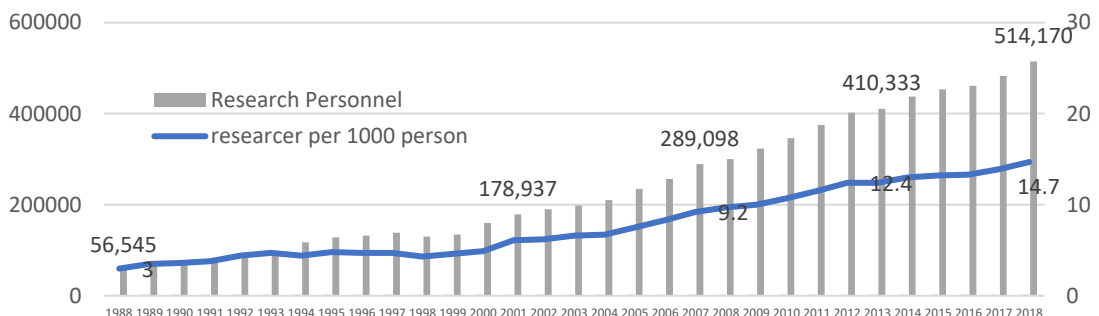
[Figure 3-2] Trend of R&D Expenditure in Korea (1976~2018)



Source: Author's own elaboration based on data taken from Ministry of Science and Technology-KISTEP, index.go.kr (as of 2020)

The Number of Research Personnel (Full-Time Equivalent) has grown, reaching 514,170 persons in 2018. It has been showing a constantly increasing trend from 178,937 in 2001 to 289,098 persons in 2007. Researchers per 1,000 persons also increased from 3 in 1989, 9.2 in 2008, and currently 14.7 persons in 2018 (Ministry of Science).

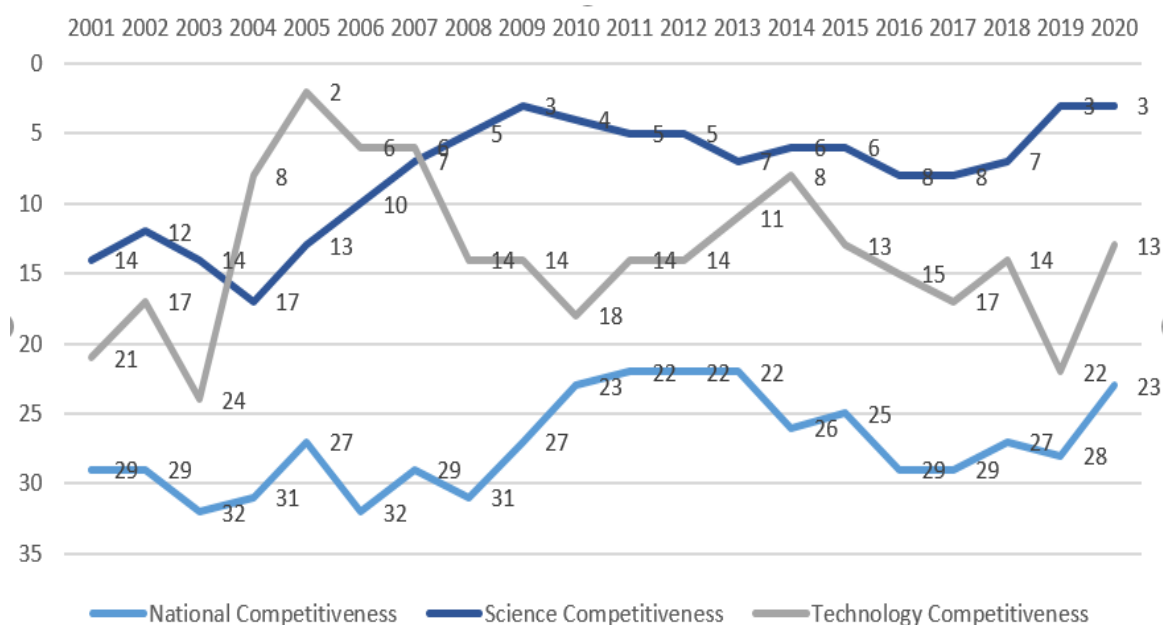
[Figure 3-3] Trend of Research Personnel in Korea (1976~2018)



Source: Author's own elaboration based on data taken from the Ministry of Science and Technology-KISTEP, index.go.kr (as of 2020); the number refers to Full-Time Equivalent (FTE) researchers

The global competitiveness of Korea is also increasing. In 2020, Korea ranked 23rd in terms of competitiveness among 63 IMD countries. Especially, it ranked high in the world science competitiveness in 2019 and 2020, placing 3rd.

[Figure 3-4] Trend of Korea's National and S&T Competitiveness (2001~2020) (unit: rank)



Source: IMD World Competitiveness (worldcompetitiveness.imd.org)

2.2 Key Features of Korea's NIS⁹

The current system of Korea's STI system began from the economic development strategy that consistently constraint STI policy with industrial development. Korea's industrial development strategy in the early stages, characterized by government intervention, strategic industry targeting, concentration of resources on strategic sectors, outward-looking development, etc., has been directly transfused into STI policy, shaping the STI system it is today. ↵

As in the case of industrial development strategy, the government intervened directly in STI in ways such as the following: ↵

(1) Creation of GRIs to help industries redress the technological deficiency of private industries; (2) Setting targets for strategic R&D and strategic resource allocation; (3) Government-initiated public-private partnership (PPP) in strategic R&D; (4) Provision of incentives for private industrial R&D and innovation; and (5) Heavy investment in advanced education on science and engineering. More notable is the Korean government's approach

to STI development based on pragmatism; Korea has been able to achieve industrialization and economic development because its STI policy has been practically oriented, such as:

⁹ We mainly referred to Chung(2017, p.104) for assessment of main features.

(1) Policies tuned to the practical objective of industrial technology development rather than the long-term advancement of scientific knowledge (Illusive scientific goals); (2) Efficient use of limited resources: resource allocation based on the principle of “selection and concentration” – Strategic focus of resources on target areas based on meritocracy, or concentration of resource allocation to those who can do better. Today, Korea is the 10th largest economy, having emerged as a major ICT power in the global market for which it owes much to the strategy above. Despite the remarkable changes in the size and structure of the Korean economy, however, many still seem to believe in the past strategy that brought about the success and stick to past strategies even when pursuing a totally new goal. The same is true in the case of STI policy. It is still not free from the legacy of the development period but hangs on to the past policy tools in dealing with entirely different goals. Briefly, Korea’s STI system today is characterized by the following:

- A. Government intervention: Despite the rapid rise of the private sectors, the influences of the government over the private sector do not seem to taper off. ↗
- B. Linkage of STI policy to national development (1) R&D investment priority set by development policy priority (2) Concentration of R&D resources on strategic areas (3) Extreme concentration of industrial R&D in large enterprises ↗
- C. Reliance on planning (1) STI policy planning: Setting numerical targets – What and how much to achieve by the time set (2) Dominance of top-down designed R&D programs ↗
- D. Industry/GRI-centered STI system: Dominance of the private industrial sector in R&D, with GRIs being the major player in public sector R&D and universities a minor player.↗

The above features are outgrowths of the past development strategy. While the strategy was very effective in maximizing growth under the severe resource constraints, it also resulted in a serious imbalance in the STI system. These included the imbalance between basic R&D research; imbalance between strategic and nonstrategic areas – ICT vs. others; imbalance between large enterprises and SMEs in technological capability; and imbalance between universities (top 20 universities grab more than 60% of university R&D funds).↗

3. Strategic Concentration of R&D Resources

3.1 Highly Advance National (HAN) Project

A notable development in the early 1990s was the growing strategic interest in R&D and innovation among policymakers in the non-S&T sectors. Even non-S&T ministries began to draw up their own long-term plans for technology development. MOST went ahead of other ministries when it launched the Highly Advance National (HAN) Project (19) aimed at developing the technologies essential for the structural transformation of the economy. In order to identify the target technologies, MOST implemented a technology foresight project in 1991 (20), the result of which was reviewed and modified by a group of experts for finalization. The expert group consisted of scientists, engineers, and businesspeople representing the government, academe, and industries. The final list of target technologies went through a process of inter-ministerial consultation for agreement and approval. It was then finalized and put into implementation. The HAN program differed fundamentally from other government R&D programs in several ways: (1) It was a large-scale, long-term program; (2) It targeted developing the strategic technologies identified through a formal process; (3) It was an inter-ministerial program reflecting national demand; and most importantly, (4) It was organized as a public-private collaborative program wherein the government and private industries combined their research funds and manpower to attain the set goal. Its target technologies included two categories: ↵

(1) Product technologies: HDTV, ISDN, micro-machines, next-generation automobiles, and new drugs.↵

(2) Fundamental technologies: Ultra-large-scale integrated circuit, next-generation nuclear reactor, ICT materials, new functional biomaterials, advanced manufacturing systems, environmental technology, and new energy technology. The HAN project was followed by a series of long-term, large-scale, public-private collaborative programs, such as the 21st-Century Frontier R&D Program (1999–2012) launched by MOST and the Next-Generation Growth Engine Development Program (2004–2008) administered by the Ministry of _

Industry and Resources. The HAN project marked an important milestone in the evolution of Korea's R&D policy by making two distinctive contributions:⁴

- It brought the strategic planning concept into public R&D, thereby linking R&D programs systematically to national development; and⁴
- By focusing on strategically chosen target technologies, it disseminated the concept of “selection and concentration” in R&D resource allocation among the R&D management agencies in both public and private sectors. The principle of “selection and concentration” stemmed from Korea's need to overcome the disadvantages of being a latecomer that had to catch up with advanced countries while being subject to severe resource constraint.⁴

The target technologies of the HAN project clearly reflected the industrial policy direction as set forth in the 6th Five-Year Socio-Economic Development Plan (1987–91) and the Five-Year Plan for the New Economy (1993–1997). The development plans emphasized the development of technological capability in electronics, machineries, and automotive industries, where Korea was deemed to have potential to establish comparative advantage over others. In particular, the Five-Year Plan for the New Economy placed high priority on the development of IT industries and technological capability in ICT. Thus, during the plan period (1993–1997), the government created the Ministry of Information and Communications (MIC) as the central agency to implement the policy and launched the Korea Information Infrastructure Initiative (KII), a ten-year program to construct an information highway. In addition, they established the ICT Promotion Fund (a joint fund of the government and ICT industries; USD 1–1.5 billion per year) to finance public-private joint R&D for ICT technology development. To coordinate all these programs and related activities, they established the Information Society Promotion Committee with the Prime Minister as its chair. Therefore, during the entire period of the 1990s, the lion's share of government R&D funds was allocated to a few strategic areas—ICT, machinery, and material technology received 60–75% of the funds of the 10 major R&D programs of the government.⁴

[Table 3-5] R&D Full Allocation of 10 Major Government Programs by Area (unit: %)

	ICT	Machine	Materials	Bio-tech	Energy	Others
1992	47.6	15.4	12.6	3.7	10.5	10.2
1993	37.4	18.8	14.1	3.6	13.9	12.2
1994	36.2	18.1	12.6	5.3	11.8	16.0
1995	37.5	13.9	11.1	5.9	12.2	19.4
1996	36.2	12.2	12.4	5.8	9.5	23.9
1997	38.0	15.0	12.4	5.2	11.3	18.1

Source: Cho, et al (1998), retaken from Chung (2017)

The strategic emphasis on ICT was reinforced around the turn of the decade. When the Korean economy was hit by the financial crisis at the end of 1997, the government resorted to ICT as a means of revitalizing the economy. They not only accelerated ICT R&D but promoted ICT startups by offering diverse subsidies and incentives that included financial assistance (for startup investment, technology development, and marketing), tax incentives (tax deduction or exemption on income from investment in startups), provision of credit guarantee, etc. (Jang, 2005). Beginning 1998, the number of new technology startups increased rapidly, reaching 11,392 in 2001. Even though the growth trend of startups reversed in the face of the global dot.com disaster in the early 2000s, it did not take long for the Korean ICT ventures to bounce back. The explosive increase of new startups in Korea around the end of the 1990s was facilitated by several factors:

- (1) Many saw new business opportunities from the dot.com boom in industrialized countries during the period 1995–2001;⁴⁾
- (2) Many researchers who lost R&D jobs because of the financial crisis of 1997 sought new opportunities in ICT, particularly in Internet-related business. Many of them set their sights on starting new ICT- and Internet-based businesses.⁴⁾
- (3) The government encouraged and promoted such movements as a means of renewing economic dynamism, offering diverse incentives for new technology-based startups. ⁴⁾

(4) The Korea Securities Dealers Automated Quotation (KOSDAQ) provided a medium for new startups to translate their business ideas into economic values.⁴⁾

3.2 National Technology Roadmap (2002)

In Korea, R&D fields were chosen according to the direction of the government's promotion; the position of science and technology as a subject for solving social problems and as a tool for economic growth has steadily improved. The national key R&D sector was selected in line with the path of economic growth fostered by government and domestic and international environmental change.

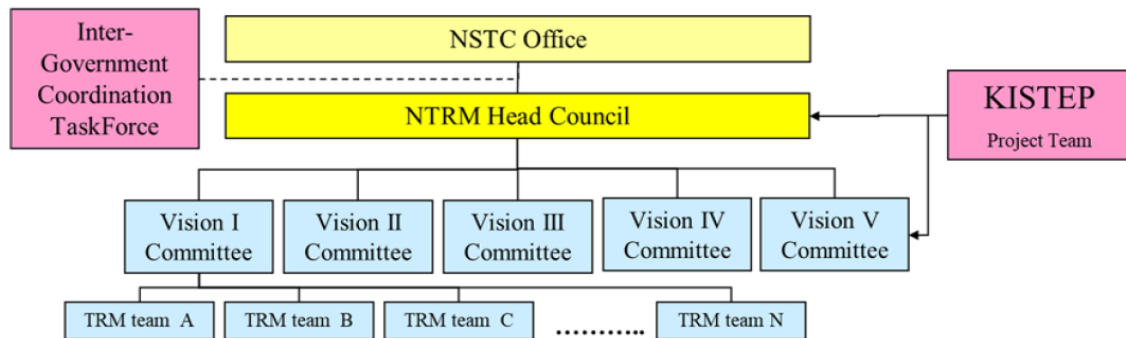
In the 1970s, research and development were promoted in the fields of machinery, metal materials, electronics, and chemicals in order to support the heavy and chemical industries strategically. In the mid-1980s, public technologies (environmental fields) and large scientific fields (marine, aerospace, and space), which are difficult for the private sector, were added to conduct research and development. Efforts were made in the 1990s to acquire original technologies and core strategic technologies for the advancement of existing industries and entry into technologically advanced countries. In the 2000s, research and technology advancement that will lead the future of society, such as future source technologies and future promising technologies, was presented and implemented inclusive of industrial technologies and public technologies.

The Ministry of Science and Technology (MOST) developed a national roadmap for technology in 2002 called the National Technology Roadmap (NTRM). The purpose of establishing the NTRM was to come up with national R&D programs more closely related to market needs (MOST and KISTEP, 2002). The main objective of the NTRM was to strengthen core technology development capabilities at the national level in preparation for future technology demand. This was a preparatory work for the efficient use of our science and technology resources and capabilities, which were at a lower level compared to advanced countries, through selection and concentration. The National Technical Guidelines consisted of a wide range of work by 17 relevant ministries and 800 experts from industry–academe–

research.

The main activities related to the establishment of NTRM have been directed by the NTRM Head Council. The Executive Committee was also set up with five vision committees, which formed the central body for the creation of the NTRM. Asif from the 5 vision committees, 74 TRM teams have been formed to build the main technology TRMs. Each TRM team was made up of about ten technology experts from the industry, universities, and research institutions. A total of 751 members of the Committee participated in the preparation of the NTRM. Existing TRMs have been integrated into the NTRM, with some changes. This process was completed within a short period (from March to December 2002), which could be attributed to the fact that there was a certain extent of previous expertise and understanding of the NTRM prior to the start of this full-scale work. The project team has worked tirelessly to ensure that all the necessary work is carried out within the current administration that was in place at the time.

[Figure 3-5] Organizational Structure for NTRM Establishment



Source: STEPI(2018¹⁰), p.25

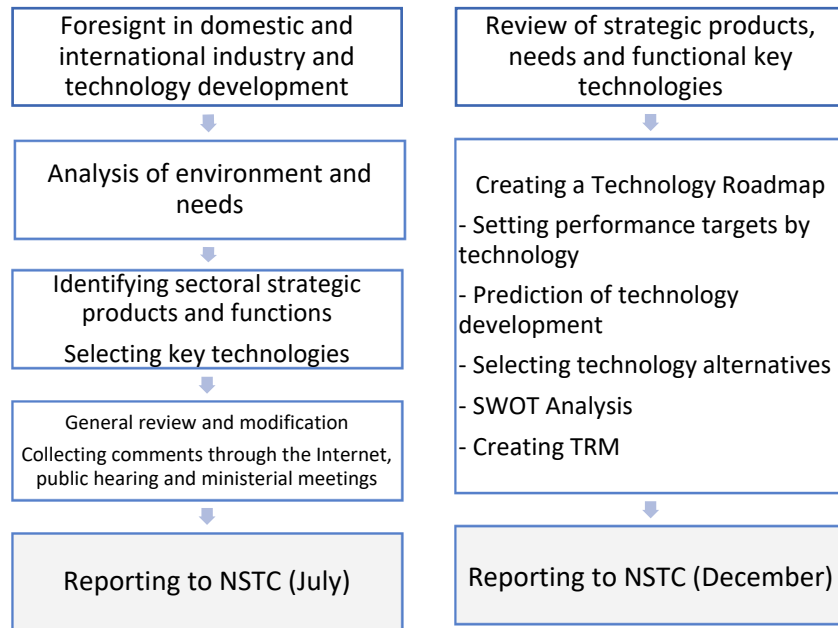
¹⁰ STEPI (2018)

According to STEPI (2018), “the NTRM activity was carried out in two phases. The first phase was to identify technologies for which the NTRM was drawn up. It explored the vision of national S&T development for the next ten years and the elements to be secured for industrial competitiveness. It forecast general technological elements as well as non-technological ones for enhancing global competitiveness by 2012 and defined strategic products or functions and key technologies required to realize those visions.”¹⁾

The strategic products or functions were identified by prioritization. A typical prioritization was made through a group decision-making process of discussion, debate, and consensus building among the domain experts. Three prioritization criteria—economic impact, strategic importance, and potential for success—were used to identify strategic products or functions., where economic impact and strategic importance were evaluated by a 5-point scale: very high (5), high (4), middle (3), low (2), and very low (1). Potential for success was measured by a 3-point scale: high (3), middle (2), and low (1). ²⁾

The second phase drew up TRMs for key technologies identified in the first phase. It derived future visions related to key technology areas and marked milestones of technology development to achieve these visions. It identified technological alternatives in each technological area, which were necessary for attaining the target technological capacity, and explained how to reach the target by presuming the process and using time coordinates. A macro TRM was prepared for each of the 49 strategic products and functions. A micro TRM was prepared and presented for each of the 99 key technologies. The NTRM includes Macro Roadmaps for strategic products/functions and detailed TRM for the chosen key technologies.³⁾

[Figure 3-6] Two-phase Process in Establishing NTRM



The strategic roadmap was prepared for each of the 30 national critical technologies. The structure of the strategic roadmap consisted of three parts: environmental change prospect and technology situation analysis; selection of key technologies; and strategic direction.

The "National Technology Roadmap" created in 2002 is a form of society that we must achieve ten years later. First, five visions were identified: 1) realization of an information-knowledge-intelligent society; 2) target of a safe life society; 3) promotion of the environment-energy frontier; 4) creation of core industrial value; and 5) improvement of national security and status. In line with this, the demand for future goods and core technologies was identified. On the basis of such, 49 strategic products and functions were selected, and 99 core technologies needed for this were obtained. A detailed map with step-by-step progress milestones was prepared for each core technology. 6) The National Technology Map later became a blueprint for the development of science and technology, embodied in the "Basic Science and Technology Plan" and the industry's R&D strategy, and it was also used to plan the development of science and technology for each field such as the National Nuclear Technology Map and to create a technology map.

almost every TRM including consequent methodologies has been tried and studied at the country level. Behind the creation of a grand TRM is the Korean government and the agencies carrying out the roles of principal and agency, respectively. One of the Korean ministries, MOTIE (Ministry of Trade, Industry, and Energy), has been in charge of industrial and technology policies and has always been passionately involved in developing new policies to cope with new presidents and emerging needs from industries. The TRM is fascinating enough to satisfy the high-level policymakers inside MOTIE, and the industry participants welcome this new way of technology evaluation methodology because they are invited to join in the process of consensus building. ↵

In the infancy stage of TRM, there was only one agency that could pursue technology evaluation research for public use. The agency, named ITEP (Institute of Technology Evaluation and Planning), had contributed to establishing a technology policy infrastructure including TRM methodology, and it had been divided into KEIT (Korea Evaluation Institute of Industrial Technology), KIAT (Korea Institute of Advancement of Technology) and KETEP (Korea Energy Technology Evaluation and Planning). Their main missions are as follows:↵

- KEIT: Evaluating public R&D projects and related planning↵
- KIAT: Developing technology and infrastructure↵
- KETEP: Promoting energy technology and innovation↵

↵

Even though these three technology agencies are under the auspices of MOTIE (Ministry of Trade, Industry, and Energy), they have been given their own roles, and there has been a clear distinction between MOTIE and the agencies in many ways, i.e., the roles of principal and agency. Developing the technology policy and TRM including the related methodology and implementation has been the main task of KIAT, but such was not monopolized by any of these agencies. In order to establish the appropriate industry areas inevitable for the priority setting of the Ministry, researchers in KIAT are constantly engaged in discussions with government officials from MOTIE on how to develop the TRM methodology and implementation plans. Once policies and plans are settled, KIAT then starts to implement the TRM process—such as basic plan, formation and operation of steering and technological

committees by areas, holding of several public hearings and, finally, publishing of TRM online and offline. Then, the results of the TRM are, by law, to be conveyed to KIET and to KETEP and materialized in the RFPs for programs and consequent projects. All evaluation processes of developing technology—from posting the RFPs (Request for Proposals) to remitting the government money, which is usually matched, to the bank account of the awardee of the project—are undertaken by KEIT. The process of TRM and technology development as mentioned above naturally leads to the question of why Korea has a separate and independent agency solely for the energy sector, KETEP. This question can be easily answered by referring to Korea's energy dependence on imports, which is 97%. Korea produces no oil or natural gas, although its economy enjoys a huge trade surplus in the oil and chemical industries.

3.3 Priority Industry-Setting Procedure by the Presidential Administration

The government set up a Next-Generation Growth Engine Selection Committee under the National Science and Technology Commission in 2003 and identified “the top 10 next-generation growth engine industries.” The government has selected intelligent robots, future vehicles, next-generation batteries, displays, next-generation semiconductors, digital TV/broadcasting, next-generation mobile communications, intelligent home networks, digital contents/SW solutions, and bio new drugs/long-term as the next-generation growth engines.⁷⁾ For the 10 selected areas, the government and private sector jointly spent a total of 2,287.8 billion won (government: 1,555 trillion won, private sector: 7,238 billion won) from 2004 to 2007, concentrating on key tasks. There was a lot of investment in the field of information technology.⁸⁾ The government intends to develop 40 strategic product lines and 153 key technologies that can generate tangible results within 10 years. In order to promote the success of the task, a special committee for the promotion of next-generation growth engines and a general working-level committee were set up under the National Science and Technology Committee and convened regularly.

In 2005, in order to systematize the budgetary adjustment and distribution of national R&D

projects from a mid- to long-term perspective, the "National Mid- to Long-Term Development Strategy for R&D" was developed jointly with related ministries. This strategy involves policy tasks from a cross-ministerial perspective, such as the investment portfolio strategy between 9 technology fields, selection of 33 advanced technologies and 57 candidates for specialization, expansion of basic and source technologies, and correlation between mid- and long-term plans. It was designed to be applied systematically to the R&D project promotion phase of each department.

In 2008, the government set up a "national R&D investment plan" to broaden the national R&D budget and improve investment productivity over the next five years. The mid- to long-term growth strategy for national R&D projects was developed in the same year, and 50 core technologies and 40 candidate technologies in 7 fields were selected and reflected on the Lee Myung-bak Administration's Science and Technology Basic Plan. In 2009, the following year, "Comprehensive Measures for Green Technology R&D" and "Key Green Technology Development and Commercialization Strategies" were prepared to support "green growth" (Committee on Green Growth, 2009). In order to forecast the future of Korean society in 2040 and to present the goals to achieve it in terms of science and technology, the "Science and Technology Future Vision and Strategy" were established, and 25 core technologies for the future were presented.

The new Lee administration has found a new growth engine in a more expanded field. It broadened the perspective from manufacturing and system technology to a practical perspective connecting innovations and industries such as the green technology industry, the hi-tech integration industry, and the high value-added service industry, covering most of the core tasks of the previous administration. As a result, the "New Growth Engine Vision and Development Strategy" were established and announced in January 2009, and they included 17 new growth engines in 3 fields. A detailed implementation plan consisting of 200 projects was announced in May 2009 along with measures by function such as technical strategy guidance, manpower training plan, and SME support plan. Between 2009 and 2012, a total of 8.7 trillion won was invested in new growth engine enterprises, and a policy loan including related funds worth 3.9 trillion won was generated. This project sponsored the

reform and enhancement of legislation and regulations for the revitalization of private investment, utilization of public demand, and pilot projects with the goal of promoting the sector in a significant way.

A total of 120 strategic technologies and 30 core national technologies have been selected in the "3rd Basic Science and Technology Strategy" developed in 2013. In 2014, 15 ministries and 205 industry-academe-research experts participated in the development of the "National Key Science and Technology Strategy Roadmap" in order to promote national key science and technology strategies effectively (PACST, 2014a). Unlike the existing National Technology Map, which centered on evaluating and planning extensive technology plans, this Strategic Roadmap comprehensively included the Technology Roadmap, practical and marketing strategies, legal and institutional reforms, and cross-ministerial linkages. In 2016, the "Government R&D mid- to long-term investment strategy (2016~2018)" was identified, and investment strategies for each of the 9 technology sectors and 58 main areas were created, and efforts were made to link the allocation and adjustment of the government's R&D budget significantly.

[Table 3-6] Key Technologies as a "Next-generation Growth Engine" by the Government

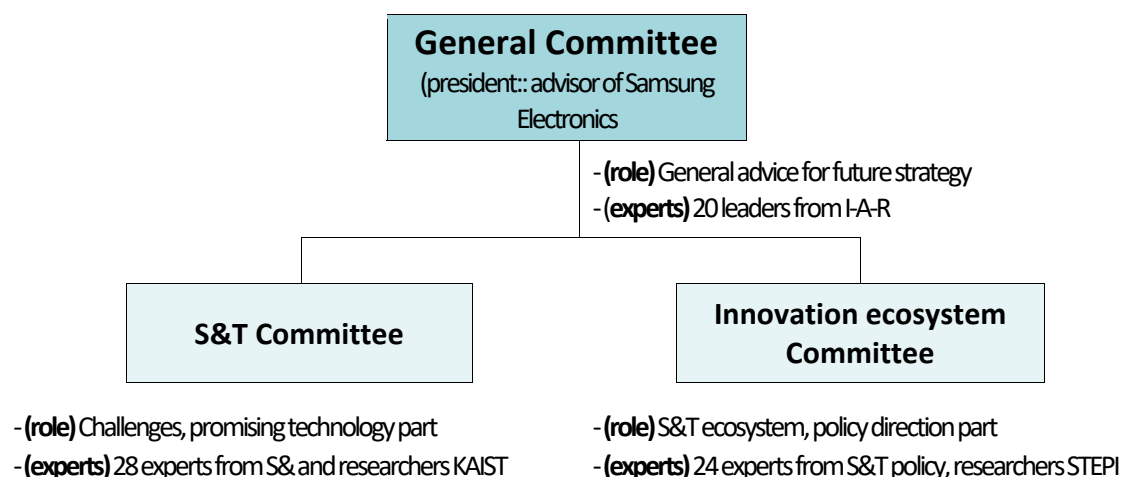
Government	Key industries and technologies	
10 next-generation growth industries	Intelligent robot, futuristic car, next-generation battery, display, next-generation semiconductor, digital TV/broadcasting, next-generation mobile communication, intelligent home network, Digital Content/SW Solution, Bio New Drug/organ	
3 categories, 17 industries	Green technology	New renewable energy, carbon reduction energy, advanced water treatment, LED application, green transport system, hi-tech green city
	Advanced Convergence Industry	Broadcasting Communication Convergence Industry, IT Convergence System, Robot Application, New materials/nano convergence, biopharmaceuticals/medical devices, high value-added food industry
	High value-added service industry	Global healthcare, global education service, green finance, contents/software, MICE/tourism

Government	Key industries and technologies	
19 future growth engine industries	Future new industry	Intelligent robot, wearable smart device, realistic content, Smart bio production system, Virtual training system
	Major industries	Smart car, deep-sea offshore plant, 5G mobile communication, vertical take-off and landing drone
	Public welfare-energy industry	Customized wellness care, renewable hybrid, disaster safety system, DC transmission and distribution system, micro power generation system
	Infrastructure industry	Convergence materials, intelligent semiconductors, IoT, Big Data, advanced material processing system

Source: Jung, et al (2008), Future Committee (2009), retaken from STEPI (2017)

The current government of Korea has published the 4th Science and Technology Basic Plan (2018) and “S&T Future Strategy 2045” (2020) as a new long-term plan. The Committee was formed, and more than 20 committee meetings were held from 2019 to set up the vision and identify the target industry in line with that. A public survey in addition to public hearing was newly included in the process.

[Figure 3-7] Structure of the Committee for “Science and Technology Future Strategy 2045”



4.

National Master Plan for Human Resources in Science and Technology (HRST)

4.1 Human Resources in Science and Technology in Korea

4.1.1 Main Feature

“The rapid growth in R&D expenditures in Korea can be explained by two factors: demand and supply. On the supply side, Korean industries have been able to meet the increasing demand for R&D and innovation because they have been financially able to do so, and also because their investments have been backed up by well-trained human resources. The government contributed to such development in several indirect ways (Chung, 2009¹¹).”⁴

Korean industries have been able to increase R&D investments at such rapid pace, thanks to the extensive pool of highly educated manpower. Considering the fact that R&D investment is more constrained by the lack of human resources than financial limitation in both developed and developing countries, we can say that Korea prepared itself well for development by investing heavily in advance in education and human resource development (Chung, 2009¹²).⁴

“The high level of educational attainment among the Korean people is attributed largely to the Confucian tradition, which places the highest value on education and scholarship. Confucianism teaches that education is a process of not just learning but also achieving moral perfection. Thus, education has served as the primary vehicle for socio-economic advancement in Korean society. The strong desire for education in Korea stems from both the Confucian culture and the consequent high returns on investments in education in the form of socio-economic advancements).”⁴

One of the most serious constraints that hindered Korea’s development efforts was the lack of high-caliber scientists and engineers. In the middle of the 1960s, Korea’s R&D manpower

remained at about 4,000 (or 1.3 per population of ten thousand), and engineers or higher-level S&T manpower accounted for less than 24% of the total (about 74,000; MOST 1968); this means that Korea at the time suffered a serious lack of high-caliber S&T manpower.

[This deficiency curtailed Korea's technological learning capacity. On the other hand, the Long-term Plan for S&T Development (1967–86) forecast that the industrial development would generate rapid increases in demand for R&D as well as industrial S&T manpower. The plan forecast the demand for R&D manpower and industrial S&T manpower to increase to 56 thousand and 240 thousand, respectively, in the 1980s, well beyond the level that Korea's S&T education system could cover.⁴³

Therefore, the government took a two-track strategy. A short-term strategy was to repatriate qualified ethnic Korean scientists and engineers from abroad. The government drew up and implemented a program offering very attractive terms to encourage the return of Korean scientists and engineers living abroad. In the late 1960s and 1970s, the program brought back over 500 scientists and engineers for short- or long- term employment at GRIs, industries, or universities (MOST 2008). Many more returned to Korea during the same period through non-government programs. The other strategy was to cultivate the necessary brains at home by reforming and strengthening the domestic system of advanced education in S&T. It was then widely agreed that creating a completely new system would be far more effective in strengthening and improving the nation's graduate education system than reforming the existing ones. For this purpose, the government created KAIS, an institution for advanced education and research in science and engineering modeled after the US system of graduate education. In order to promote KAIS as a model for graduate institutions in science and engineering, they set the following guidelines in operating the institution: (1) Recruit top students by offering incentives such as exemption from military duties and provision of financial assistance that covers tuition, room and board, and other incidentals; (2) Recruit top-notch professors by providing top-level salaries, housing, research funds, and so on; (3) Adopt the US graduate system of linking education to research; (4) Ensure independence and autonomy in education and research by putting KAIS under MOST as a way of shielding it from the interventions of the Ministry of Education; (5) Place higher priority on nurturing engineers for industries. In other words, the objective of KAIS

was not just producing high-quality scientists and engineers, but also stimulating the reform of the existing graduate system.

4.1.2 Overview of HRST in Korea

The National Master Plan for Human Resources in Science and Technology is rooted in the human resources development policy in science and technology (“HRST policy”). This policy was developed in tandem with the government-initiated economic development, and its characteristic can be differentiated before and after enacting “the Special Act on Supporting Scientists and Engineers for Strengthening National Science and Technology Competitiveness” (KDI, 2016).

In the 1960s, the HRST strategy centered on encouraging and securing qualified human capital required to boost the light industry, since the Korean government selected the sector as the top promotion priority and the key industry under the first and second Five-Year Economic Development Plans. The HRST program in the 1970s continued with the success of the economic development strategy in the 1960s. Moreover, the acceleration of the industrial structure has caused the driving force of economic improvement to be replaced by the heavy industry, which has seen the government adopt a strategy aimed at developing scientists and engineers for the heavy industry.

In the 1980s, the government focused on establishing a system for developing highly skilled scientists. This simultaneously created the necessity of stabilizing the heavy industry, including textiles, petrochemicals, electronics, machines, and shipbuilding, which had been fostered as a basic industry from the 1970s. Moreover, the technology-intensive hi-tech industry affecting national competitiveness in the future, such as the information technology and energy resources industry, was highlighted. In the 1990s, the HRST policy no longer supported industry development because of high economic and industrial growth. Instead, it concentrated on fostering highly skilled S&T human resources with competitiveness for technology innovation and development of fundamental technology as the driving forces of industrial development.

In the early 2000s, the Korean government changed the policy direction from supply-oriented to demand-oriented because of the private sector's increased demand for S&T manpower and rapid changes in science and technology. In April 2004, upon enacting, proclaiming, and enforcing the Special Act on Supporting Scientists and Engineers for Strengthening National Science and Technology Competitiveness, the government began implementing the HRST policy in earnest. In particular, the first master plan established in 2004 was the most suitable plan to advise and improve the status of Korea's S&T human resources. Therefore, we mainly address this master plan in this chapter. The first master plan established in 2004 consists of five major sections.

The first section pertains to system innovation of university education in science and engineering. This part includes university specialization in science and engineering, promotion of competition between and within universities, and quality improvement of human resources in science and engineering by reinventing the education process, etc. The second section involves developing core R&D human resources. This part includes various programs to improve highly skilled manpower in universities. The third section pertains to cultivating demand-driven human resources. This part covers strengthening the foundation for promoting industry-university/government, enhancing continuing education and reeducation of manpower in science and engineering, promoting the industrialization of developed technology, etc. The fourth section pertains to welfare support for scientists and engineers. This part outlines policies for promoting support for science and engineering, such as forming a continuing research environment, supporting welfare improvement, expanding the entry of official position, promoting employment, etc. The last section discusses infrastructure support for science and engineering manpower. This part consists of promoting understanding and widening the basis for activity related to science and engineering and establishing infrastructure for providing information on human resources in S&T, etc.

The first master plan focuses on fostering scientists and engineers in universities. In other words, the plan concentrates on increasing support or changing the system to strengthen the capacity or role of a university. It also boosts efforts to make scientists and engineers an

appealing job and to maintain diverse statistics or information utilized for policy.

On the other hand, the second master plan focuses on nurturing creative S&T human resources based on the first project's performance. Along with fostering scientists and engineers, the plan suggests strengthening future education, STEAM (Science, Technology, Engineering, Arts, and Mathematics), in elementary and secondary education as one of the main projects. Cultivation of people with STEAM literacy, special education for the gifted, and linkage between education and career, etc., are the plan's other major fields. The second master plan sets the goal of Specialization of the University and Reinforcement of Global Research Capacity for the university system. It also suggests a selection and concentration strategy for investing in major universities and specialization of local universities. When it comes to the research institutes, the second master plan aims to strengthen the public research institutes' functions to foster human resources by reinforcing the research immersion environment for researchers, expanding the participation in education through the use of infrastructure and research facilities owned by the institutes, and concentrating on the function of nurturing manpower with executive ability. This plan strengthens support for corporations that contribute to the research capacity and development of human resources by improving the ability to react to the demand for human resources in the research sector by promoting education and cultivating corporations with good research capacity. The second master plan also emphasizes the use of foreign scientists and engineers and increasing support for female scientists and engineers. Encouraging the efficient utilization of existing manpower is what sets it apart from the first master plan.⁴⁾

[Table 3-7] Paradigm Shift of Human Resource Policy

Period	Policy paradigm	Main features
1960s	Capacity building for light industry technician	- 1st 5-year technology promotion plan - Establishment of Government Research Institute, repatriating qualified researchers abroad
1970s	Capacity building for heavy chemical industry technician	- Long-term human resource plan - Establishment of KAIS

Period	Policy paradigm	Main features
1980s	Expansion of University education	- Comprehensive development strategy for S&T personnel - Establishment of University of Science and Technology, POSTECH
1990s	Enforcement of high-quality education in the graduate school (R&D)	- Concentrated support program for engineering schools as well as graduate school
2000s	Quality improvement of R&D personnel Resolving the avoidance culture of science and engineering among students Regional development of HR	- Support plan for science and engineering personnel - Science and technology basic plan (577 initiatives)

Source: Lee, et al (2008), Research for future development of HRST (KISTEP)

4.1.3 University Evaluation and Accreditation

In 1982, the Korean Council for University Education (KCUE) established the university institutional accreditation (the name of the institution was changed to the Institutional Accreditation System in 1994) in order to evaluate the quality of universities systematically. From 2010, the Korean University Accreditation Institute (KUAI), a KCUE-affiliated organization, has been implementing the accreditation for higher education institutions. The accreditation institute guarantees the quality of a university and examines whether each institution satisfies the minimum requirements through the university institutional accreditation. The criteria consist of six categories: (1) mission and development plans; (2) education; (3) institutional community; (4) education facilities; (5) financial management; and (6) community service.

KUAI allows the university to submit first a self-assessment report on the evaluation categories. Simultaneously, it organizes an evaluation group for objective accreditation and implements systematic assessment through site visit; this allows the evaluation to gain public trust. After the report is finished, the evaluation group reviews the result, determines

the final outcome through a verification process, and publicizes it. The accreditation is given to an institution depending on whether it met the evaluation criteria or not. There are four types of accreditation: full accreditation, provisional accreditation, deferred accreditation, and no accreditation. An institution with full accreditation can maintain the accreditation for five years, and institutions with provisional accreditation must be re-evaluated after a period of time. Through this process, universities are encouraged to improve their capacity to be accredited in the future. By participating in the evaluation, the university can gain not only public confidence but also an opportunity to self-analyze their performance and improve according to the assessment. Furthermore, in case of good performance, the university may receive the government's administrative and financial support in the future. Until 2015, 173 out of 193 universities applied for accreditation (KUAI, 2015).

4.1.4 Policy for Labor Supply and Demand Imbalance

The Ministry of Education improved the mid- and long-term plan for the labor supply and demand in collaboration with the Ministry of Employment and Labor by breaking down the 6 academic levels into 35 major levels so that the university could adjust their capacities to prepare for the future. Furthermore, the Ministry of Education supported the university structure reform by implementing the Program for Industry Needs Matched Education (PRIME) to encourage universities to foster human resources of promising industries.

4.1.5 Accreditation for Engineering Education (Improvement of Education Quality)

The Accreditation Board for Engineering Education of Korea (ABEEK) has been implementing and managing the Accreditation for Engineering Education since 2001 to develop human resources for the practical engineering field by providing standards and guidelines for engineering education. This system requires bachelor's degree students to complete the program certified by ABEEK as part of their curricula. Through this, students taking the program can be exempted from the preliminary screening, and they can have a competitive advantage over students with a general bachelor's degree in engineering when entering the

domestic or overseas labor market. Furthermore, this system evaluates whether the engineering education program offered by educational institutions meets the system's criteria. It also provides the guidelines for accreditation standards to help each institution improve education quality in the future. In addition to Korea, other nations in the Washington Convention promote this program actively with the goal of nurturing outstanding human resources for the engineering field. ABEEK receives an application from the participating institutions every year and officially announces its evaluation plan to the relevant university when beginning the assessment. After choosing a target university, ABEEK organizes the evaluation team for each program consisting of one or two evaluation members and one leader and sends a list of evaluation committee candidates to the university. If there is any disagreement on the list of candidates, each university may request a change of evaluation members by giving an acceptable reason. Through this process, the committee is finally determined. For the review, each university should write and submit a self-evaluation report based on ABEEK's criteria and standards. The evaluation team reviews the report of each university and visits them to receive a more specific explanation on the self-evaluation. The committee submits the pre-evaluation report with qualitative and quantitative evaluations to ABEEK. This institution reviews internally the pre-evaluation report and submits it to the university for review. The university can raise objections to the report. After gathering all opinions, ABEEK notifies the relevant ministries of the accreditation decision. The whole process takes about a year.

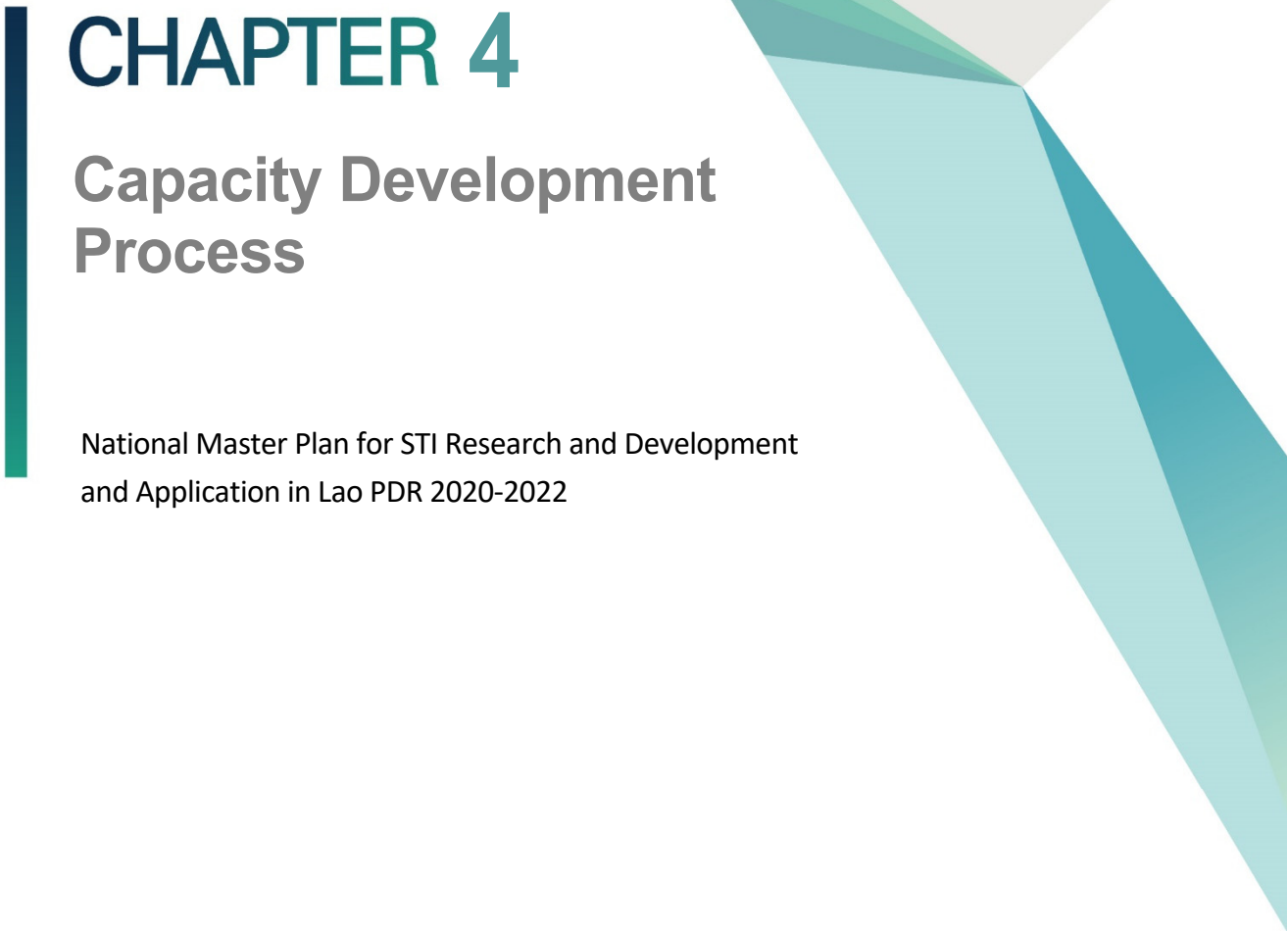
4.2 Case Study: Establishment of Korea Advanced Institute of Science and Technology (KAIST)

It was crucial to secure skilled human resources for Korea to expand its national capacity and economic capability strategically. During the 1960s and 1970s, Korea focused on the light to heavy industry with an export-oriented strategy wherein low-level technicians and cheap labor force were the main economic drivers. In order to enhance its national capacity, however, high-quality human resources who could contribute to national strategic technical advancement and industrial development were needed. These high-quality personnel would assist Korea's mid- to long-term development in science and technology. With this in

mind, the Korea Advanced Institute of Science (KAIS) was established in 1971; in 1989, KAIST was separated from KAIS as a full-fledged university. KAIST initially started as an institute offering master's and PhD degrees but now provides education from the bachelor's level.

As the development of science and technology was one of the key strategies of Korea's economic development, many special legal frameworks were established to support the science and technology institute, one of which was the 1980 Korea Science and Technology Law. In order to establish KAIST, the Korea Science and Technology Law was enacted to educate future high-level science and technology talents who could assist Korea's industrial development through research and development in the field. Such initial governmental support allowed the institute to have a more flexible governance structure. In addition, as the institute was categorized as a specialized university under the Ministry of Science and Technology (MOST) rather than the Ministry of Education, the institute was, in a sense, more autonomous. MOST provided government support, development, and supervision of all activities of the institution. As the institute is a government-driven organization with clear alignment with government policies to assist industrial development, its industry-university cooperation is stronger than that of other public universities (Kim and Hong, 2013). In the early 1980s, KAIST built the Technology Business Incubator/Technology Innovation Center for the very first time in the country to promote technical transfer and incubate new businesses. KAIST's technical transfer activities currently produce approximately KRW 27.9 billion in profit.

One of the comparative advantages of the institute is its funding structure. Despite the fact that KAIST is a public university, its funding structure differs from general public university systems in Korea. The institute is not categorized as a public university but as a public institution. As the special law states, the institute is to be funded by contributions under the law for the management of public institutions. All funding related to the institution's establishment, research, and management shall come directly from the government. Thus, KAIST receives a lump sum directly from the government. KAIST is also allowed to conduct profitable activities as long as they do not affect the research and teaching activities⁴⁾



CHAPTER 4

Capacity Development Process

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022

Chapter 4. Capacity Development Process

1. Preparations for the First-year Project

1.1 Preparatory Meetings

1.1.1 Setting the Outline of the Project Through Email

Since it is the first year of the Official ODA Program with Lao PDR, the STEPI team contacted the research team in the Ministry of Science and Technology (MOST) of Lao PDR to discuss the expected yearly schedule and activities throughout the year based on the information negotiated in the PCPs. Normally, the business trip to the counterpart country was planned in March to start the kick-off meetings and fieldwork. From February, however, the unexpected COVID-19 situation made it difficult to make a detailed plan. In the meantime, the research team in Lao PDR started telecommuting (until 18 May) and communicated with each other via email, deciding to hold an online kick-off meeting when the telecommuting is over.

1.1.2 Sharing Ideas for the Work Scope

Due to the extended telecommuting in the COVID-19 situation in Lao PDR and Republic of Korea, the scope of the project and the tentative schedule of activities were discussed through e-mail. The discussions of the work scope include the yearly schedule and activities, budget, and contract information for local consultants. The collection of the data from Lao PDR includes baseline data such as legal and institutional environment, human resources, current status of funding for scientific and technological innovation, R&D capabilities, national standard system, STI information management system, recognition of scientific and

technological innovation, and culture. The work scope of the project was exchanged through document via email, consisting of the expected process and output of the project, analysis based on local investigation results and presentation of Korea's experience-sharing direction focusing on the system and process of establishing a basic science and technology strategy.

K-Innovation LAO PDR 2020

I. Project Title

National Master Plan for STI Research and Development and Application in Lao PDR

II. Duration

April 2020 ~ November 2020

III. Work Scopes

1. [Research in Laos] Research on the Current STI R&D Environment in Laos

- Local consultants/research institutes/universities in Laos.
- 2(two) months contract based.
- Details on research will be followed when contract is made.

1) Institutional Settings

- Reviews on the existing S&T Policy and Plan
- Legal Framework
- Institutional Basis
- Governance Structure for STI policy
 - ☞ Central government bodies
 - ☞ Government-sponsored research institutes
 - ☞ Local governments

2) Human Resources

- General Education system
- STI Educational Institution
- HRD Strategy for STI
- Current status of STI HR

3) Financial Resources

- R&D Expenditure

- FDI⁴⁾
- Other financial resources⁴⁾

4) R&D Capacity⁴⁾

- R&D in business and industry⁴⁾
- Patents and publish⁴⁾

5) Other Resources⁴⁾

- National Standards⁴⁾
- STI Information Management⁴⁾

6) STI Awareness and Culture⁴⁾

4)

2. [Research & Analysis in Laos and Korea] Assessment and Analysis⁴⁾

- Based on the result of 1, Korean experts and Lao experts will work together⁴⁾

1) International Comparative Analysis⁴⁾

2) S.W.O.T Analysis⁴⁾

3) Needs Assessment⁴⁾

- In-depth interview, Survey etc.⁴⁾

4) Sub-result: Priorities to tackle⁴⁾

4)

3. [Research in Korea] S&T Basic Plan in Korea: Process of the Establishment and Contents⁴⁾

- Korean experts⁴⁾

1) Overview: History and Achievement of STI Policy in Korea⁴⁾

2) STI Governance in Korea: Focusing on the establishment of S&T Basic Plan⁴⁾

3) Contents and Action Plan⁴⁾

4) Lessons from Korea⁴⁾

4)

4. [Draft in Korea] Guideline to establish STI R&D Masterplan for Laos⁴⁾

- Based on the result of 1~3, Korean experts and Lao experts will work together⁴⁾

1) Vision and Objectives⁴⁾

2) Action Plans⁴⁾

5. [Training Program in Korea]

- One week-long, airfare/accommodation will be provided
- Participants: 10-15 mid-level officials and/or STI experts

6. [Outreach Seminar in Laos]

- Participants: 100+ STI experts
- One-day seminar + 2~3 technical workshops upon request

1.1.3 Research Topics for Local Consultants

Since the Korean research team could not visit Lao PDR due to overseas travel restrictions caused by the COVID-19 situation, collecting all the required data through local consultants as a contingency plan was critically significant to continue the project. The Korean research team offered to make contracts with three local consultants and sent the work scope to three local consultants who were recommended by the Ministry of Science and Technology of Lao PDR. The research topics were divided into three parts:

The first topic, “Systems and other STI R&D environments,” presents key survey questions on the national development strategies, national science and technology/ R&D strategies, development of science and technology policies and plans, general legal system and science and engineering legal system, STI policy governance (by subject), IP, certification and national standards systems, and STI information management system.

Research Topic 1

Institutional Settings and Other Environment of Laos' STI R&D

I. National Development Strategy

1. National Economic/Social Development Strategy (the latest five-year plan and so on)

- Provide the document in English (if available)
- (if not available) summarize it briefly in English and provide the original one
- Who/which institution(s) initiated to establish and wrote the Strategy?
- How to implement the Strategy?
- Focusing areas in the Strategy (esp. compared to the previous ones if any)

2. National S&T/R&D Strategy

- Provide the document in English (if available)
- (if not available) summarize it briefly in English and provide the original one
- Who/which institution(s) initiated to establish and wrote the Strategy?
- How to implement the Strategy?
- Focusing areas in the Strategy (esp. compared to the previous ones if any)

II. History of STI Policy and Plans

- Chronically describe the policies
- Describe the background (economic, political, industrial environment)
- Provide the document in English (if available)
- (if not available) summarize it briefly in English and provide the original one
- Who/which institution(s) initiated to establish and wrote the Strategy?
- How to implement the policies and/or plans?
- Focusing areas in the policies (esp. compared to the previous ones if any)

III. Legal Framework

1. General Legal Structure

- Provide the general ideas on the legal system in Laos (from Constitution to ordinance or other lower level of legal system)
- How do you describe the Legislation Process in Laos?

2. STI, R&D related legal framework⁴

- Describe the legal system related to STI, R&D activities in Laos (from the highest law to ordinance or other lower level of legal system)⁴
- How do you describe the Legislation Process in Laos?⁴

IV. Governance Structure for STI policy⁴

1. Government Structure⁴

- Visualize the structure ⁴
- Provide the general ideas to understand the government structure in Laos (i.e. cabinet, 18 ministries, provincial governments, committees, councils etc.)⁴

2. STI related Central government bodies⁴

- Visualize the structure ⁴
- Who are the main actors in the decision making process? What are their roles and responsibilities?⁴
- Who are the main actors in the implementing process? What are their roles and responsibilities?⁴
- Organizational structure i.e. personnel, status⁴

3. Local governments⁴

- Visualize the structure ⁴
- How do we define the relationship between the central STI government bodies and local governments?⁴
- Who are the main actors in the decision making process? What are their roles and responsibilities?⁴
- Who are the main actors in the implementing process? What are their roles and responsibilities?⁴
- Organizational structure i.e. personnel, status⁴

4. Government-sponsored research (R&D) institutes⁴

- How many government-sponsored research institutes (GRIs)? ⁴
- Provide the list of GRIs and their research areas, homepages (if available)⁴
- Provide details of each GRI i.e. number of researchers, degree (MIS, PhD etc.), gender, etc.⁴

5. Universities/colleges⁴

- How many universities/colleges have S&T and/or engineering major? ↵
- Provide the list of universities/colleges and their majors, homepages (if available)↵
- Provide details of each university/college i.e. number of faculties, degree (MIS, PhD etc.), gender, students (gender), number of graduates, degree programs etc.↵

6. Other related institutions (Academy, vocational institutions etc.)↵

- How many other related institutions? What are they?↵
- Provide details of major institutions ↵

V. Institutional Basis↵

1. Intellectual Property Right↵

- Who/which institutions are in charge of IPR i.e. copyright, license, and so on? Role and functions↵
- Describe details on the institutions/organizations (Structure of organization, legal status, composition of employees and so on)↵
- Provide each legal framework (history of establishment and summary of the law)↵

2. Certification system↵

- How many national certifications are there? What are they?↵
- Who/which institutions are in charge of certification?↵
- Describe details on the institutions/organizations (Structure of organization, legal status, composition of employees and so on)↵
- Provide each legal framework on certification system (history of establishment and summary of the law)↵

VI. National Standards System↵

- Who/which institutions are in charge of national standards? Role and functions (for example: focal point of ASEAN Consultative Committee on Standards and Quality)↵
- Describe details on the institutions/organizations (Structure of organization, legal status, composition of employees and so on)↵
- Provide each legal framework on National Standards System (history of establishment and summary of the law)↵

VII. STI Information Management⁴⁾

- Are there any STI and/or R&D information system in the national or regional level? If so, how many? ⁴⁾
- Who/which institutions are in charge of managing STI related information? Role and functions ⁴⁾
- Describe details on the institutions/organizations (Structure of organization, legal status, composition of employees and so on)⁴⁾
- Provide each legal framework (history of establishment and summary of the law)⁴⁾

⁴⁾

⁴⁾

Important!

A. Data: latest available 5 years i.e. 2015 ~ 2019 or 2014 ~ 2018⁴⁾

B. Structure: visualize as much as possible⁴⁾

C. History: chronically describe in details as much as possible⁴⁾

D. Issues in the box should be included throughout the report ⁴⁾

E. Overall implications and insight should be included.⁴⁾

F. Proper references needed⁴⁾

G. Interview can be conducted (in this case, consult with Ms. An in advance)⁴⁾

⁴⁾

..

The second part of the research topic covers the “Human resources and STI culture” including systematic education system, STEM education system, manpower training strategy, science and engineering education environment, STI/R&D workforce status, and awareness and cultural activities related to science and technology and engineering (International Network). Present key survey questions were divided into associations, public activities, etc.

Guideline for Research Topic 2

Human Resources and STI Culture in Laos

I. General Education

1. Educational System

- Provide details of Laos' education system. See the attached example for Korean educational system in the next page
- Compulsory education? Until which grade?
- Who/which ministries or institution(s) are in charge of the national educational system?
- How many students are enrolled and graduated (by gender)? How many primary schools, middle schools, high schools? Number of teachers? Student: Teacher ratio
- How many students enter the tertiary education (universities/colleges)? Percentage among the high school graduates. Provide five-year trend data if possible.
- Requirement for the college entrance
- Issues and concerns on educational system in Laos

2. STEM Education

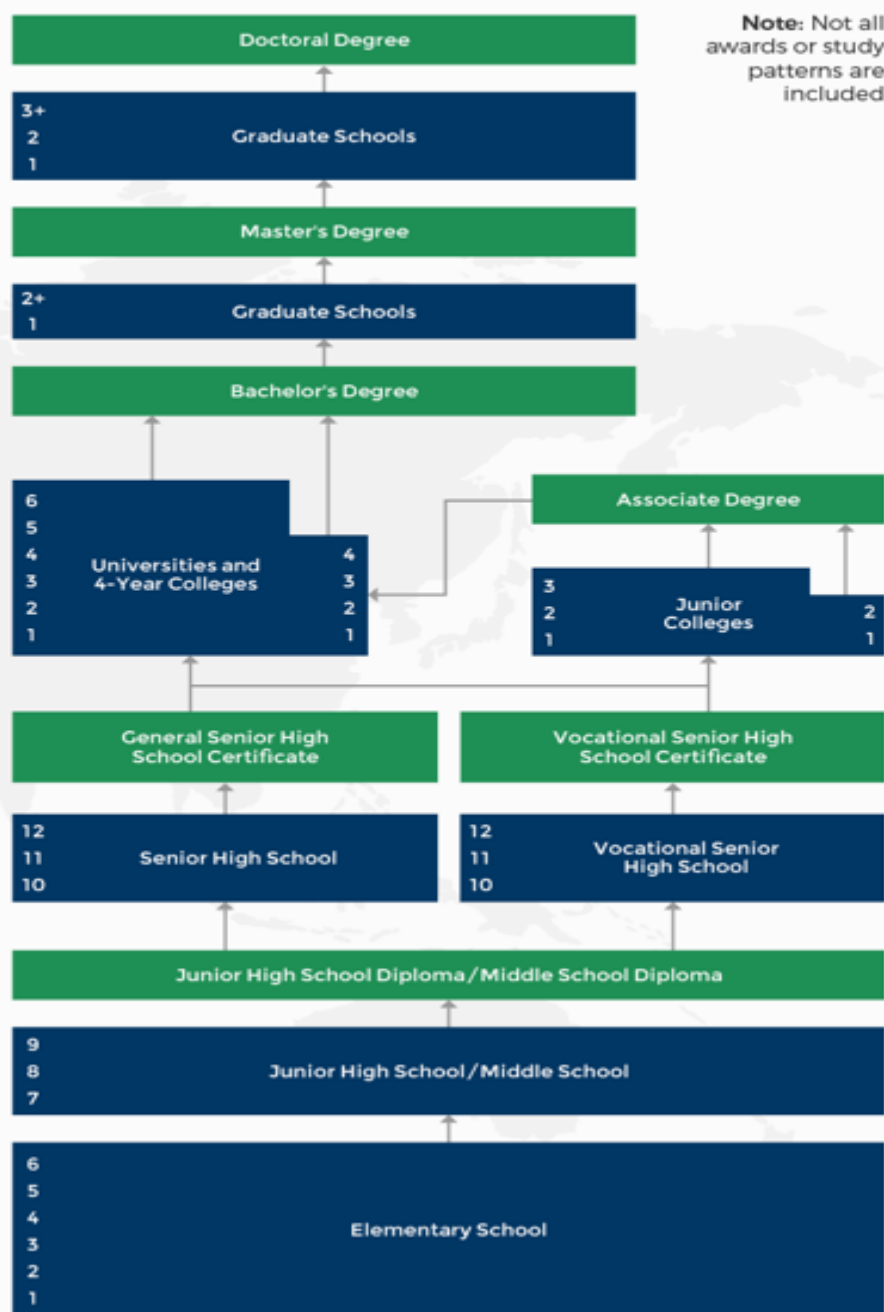
- Provide the standard curriculum and minimum requirement of each grade i.e. how many hours do students take the classes? What subjects are students taking? Etc.
- How many hours do students take the courses related to science, technology, engineering and math (STEM) in each grade?
- What subjects related STEM do schools provide for the students? For example: basic physics in grade 9 and 10, intermediate physics in grade 11, advanced physics in grade 12
- Do you have special tracks for the gifted in STEM?

3. Technical and Vocational Education Training

- How many TVET institutions do you have in Laos? Enrollment duration?
- How many attendees in the institutions above?
- Which areas are TVET institutions focusing i.e. electronics, cosmetics?
- Who/which ministries or institution(s) are in charge of the TVET?

[Educational System in Korea]

EDUCATIONAL SYSTEM



Source: NOOSR

↩

II. Human Resource Development Strategy⁴

- Do you have a national strategy for human resource development or education?⁴
- (if so) Provide the document in English (if available)⁴
- (if not available) summarize it briefly in English and provide the original one⁴
- Does the document contain STI human resources related contents specifically? Provide the details. ⁴
- Who/which institution(s) initiated to establish and wrote the Strategy?⁴
- How to implement the Strategy?⁴
- Focusing areas in the Strategy (esp. compared to the previous ones if any)⁴
- (if not) Does the government have any plan to establish such kind?⁴

III. S&T/Engineering Educational Environment ⁴

1. Higher Educational Institutions⁴

- How many high education institutions in Laos? How many are the public institutions and how many are the private institutions?⁴
- *HIEs: Institutions which can provide the degree i.e. BA. BS. MA. MIS, MS, PhD⁴
- How many students are enrolled? ⁴
- Among HIEs, how many of them have S&T/engineering department? Provide the names of the institutions and their departments. ⁴
- How many students are majoring S&T/engineering? (by gender if possible)⁴
- How many students are in the doctoral program?⁴
- How many HIEs are there having Laboratory facilities? Describe the facilities.⁴

2. S&T Learning and Teaching Capacity⁴

- Provide two-three curricula of S&T/Engineering departments in major universities/colleges if available⁴
- [Instructors/Professors] ⁴
 - Do you have enough professors and assistants to instruct the students?⁴
 - What are the major concerns for the professors in the S&T/engineering department? i.e. low capacity of students, lack of research infrastructure, burden of teaching and researching, too many classes, too many students, salary, social status and so on⁴
- [Students]⁴
 - Are they capable to follow up the advanced courses that the department provides?⁴
 - What are the major concerns for S&T/engineering majoring students? i.e. tuition, scholarship opportunity, job after the school, social status and so on⁴

IV. Current Status of STI/R&D professionals

1. Researchers and Technicians

- Number of researchers and technicians (as latest data available as possible)
- Are they employed in public sectors, private sectors, or academia?
- Research fields, level of degrees

2. Professors (select two most well-known universities/colleges, please consult with Ms. An in advance)

- Provide the detailed information on the faculty members as a table below (as much as possible)
- Gender, age range (30s, 40s, 50s, 60+), full time or adjunct and so on

[Faculty members' Information]

Department	Doctoral Degree	Master's Degree	Bachelor's Degree	Total
Biology				
Chemistry				
Computer Sciences				
Electronic Engineering				
Information Technology				
Mathematics				
Total				

V. S&T, Engineering awareness and culture

1. International S&T Network

- Please describe S&T/ICT/Engineering related relationship between the government of Laos and ASEAN
- Please describe S&T/ICT/Engineering related relationship between the government of Laos and other international organizations
- Please provide the list of the international and national conferences/seminars/meetings related to S&T/ICT/Engineering in the last three years and explain each event briefly.
- Is there any memorandum on S&T/ICT/Engineering between the government of Laos and other countries? If so, provide the information.

2. S&T/Engineers' Associations⁴⁾

- Is there any S&T researchers' and/or engineers' association in Laos? Please provide the information with the contact.⁴⁾
- Who are the members what are the main roles? What is the relationship with the government?⁴⁾

3. Public Activities⁴⁾

- Is there any activity to promote S&T in public i.e. Day of Science, national competition, Olympiad and so on? If so, provide the information.⁴⁾

Important!

- A. Data: latest available 5 years i.e. 2015 ~ 2019 or 2014 ~ 2018⁴⁾
- B. Structure: visualize as much as possible⁴⁾
- C. History: chronically describe in details as much as possible⁴⁾
- D. Issues in the box should be included throughout the report ⁴⁾
- E. Overall implications and insight should be included.⁴⁾
- F. Proper references needed⁴⁾
- G. Interview can be conducted (in this case, consult with Ms. An in advance)⁴⁾

Research topic 3, “R&D capabilities, industry, and financial resources,” presents the key survey questions including the government’s general budget and budget system, government R&D spending, private-sector R&D spending, other R&D-related financial status, patents, science and technology journals, and other ICT infrastructure, industrial development strategies, industrial structure changes, import and export, and FDI status.

Guideline for Research Topic 3 ⁴⁾	
R&D Capacity, Industry, and Culture ⁴⁾	
I.	R&D Expenditure⁴⁾
1.	General government budget ⁴⁾
	● Provide the last five-years’ government budget (including total budget, grants, revenue, sector specific allocation) ⁴⁾
2.	Budget System ⁴⁾
	● Visualize the budgeting process and structure (focusing on the main actors in the decision making process, who decides what and so on) ⁴⁾
3.	Government R&D Expenditure ⁴⁾
	● Provide the last five-years’ government R&D expenditure (gross) ⁴⁾
	● Do you have a specific budget category for national R&D? ⁴⁾
	● Do you have a national R&D Program/project? Specify them if possible ⁴⁾
4.	Private R&D Expenditure ⁴⁾
	● Provide the last five-years’ government R&D expenditure ⁴⁾
	● Who are the main actors in private R&D spending? (provide top3 global companies in Laos and tops 3 local companies) ⁴⁾
	● Any cases of University-industry joint research? ⁴⁾
5.	Other R&D related Financial Sources ⁴⁾
	● Do you have any other S&T related fund such as ASEAN Science Fund? ⁴⁾
	● Provide the details for Laos’ participation with ASEAN Science Fund ⁴⁾
	● Provide the last five-years’ S&T/ICT related Official Development Assistance in Laos. Introduce major ODA projects and describe how they have worked ⁴⁾
II.	R&D Capacity and Achievement⁴⁾
1.	Patent ⁴⁾
	● Provide the last five-years’ patent data (by origin: resident/non-resident; application/approval, areas if possible) ⁴⁾
	● Provide the last five-years’ trademark and industrial design data (by origin: resident/non-resident; application/approval) ⁴⁾

2. Scientific Journals and Articles⁴

- Provide the list of scientific journals regularly published in Laos⁴
- How many S&T /engineering articles and/or research papers are published?⁴

3. General Innovation Infrastructure and Environment⁴

- ICT Access rate, Use of ICT, government's online service⁴
- Other indicators that shows innovation capacity of Laos i.e. mobile app creation etc.⁴

III. Industrial Development⁴1. National Strategy⁴

- Provide the document in English (if available)⁴
- (if not available) summarize it briefly in English and provide the original one⁴
- Who/which institution(s) initiated to establish and wrote the Strategy?⁴
- How to implement the Strategy?⁴
- Focusing areas in the Strategy (esp. compared to the previous ones if any)⁴

2. Changes of Industrial Structure (Latest 5 years)⁴3. Export/Import (Latest 5 years)⁴

- Include high- and medium-high technology manufacturing import and export as % of total trade⁴

4. Foreign Direct Investment (Latest 5 years)⁴

- Include high- and medium-high technology manufacturing import and export as % of total trade⁴

IV. S&T, Engineering awareness and culture⁴1. International S&T Network⁴

- Please describe S&T/ICT/Engineering related relationship between the government of Laos and ASEAN⁴⁾
- Please describe S&T/ICT/Engineering related relationship between the government of Laos and other international organizations⁴⁾
- Please provide the list of the international and national conferences/seminars/meetings related to S&T/ICT/Engineering in the last three years and explain each event briefly.⁴⁾
- Is there any memorandum on S&T/ICT/Engineering between the government of Laos and other countries? If so, provide the information.⁴⁾

2. S&T/Engineers' Associations⁴⁾

- Is there any S&T researchers' and/or engineers' association in Laos? Please provide the information with the contact.⁴⁾
- Who are the members what are the main roles? What is the relationship with the government?⁴⁾

3. Public Activities⁴⁾

- Is there any activity to promote S&T in public i.e. Day of Science, national competition, Olympiad and so on? If so, provide the information.⁴⁾

Important!

- A. Data: latest available 5 years i.e. 2015 ~ 2019 or 2014 ~ 2018⁴⁾
- B. Structure: visualize as much as possible⁴⁾
- C. History: chronically describe in details as much as possible⁴⁾
- D. Issues in the box should be included throughout the report ⁴⁾
- E. Overall implications and insight should be included.⁴⁾
- F. Proper references needed⁴⁾
- G. Interview can be conducted (in this case, consult with Ms. An in advance)⁴⁾

2. Online Kick-off Meeting

2.1 Overview

2.1.1 Objectives

The purpose of the kick-off meeting is to discuss and share ideas on the project and for knowledge sharing. The kick-off meeting involves project counterparts, MOST, and STEPI to discuss the project purpose, activities, and expected outputs. Various stakeholders including government officials from the relevant ministries and the directors of the research institutions were invited to the event to build consensus on the importance of developing the National Master Plan for STI Research and Development and Application in Lao PDR. This event will identify policy demands for the main institutions and ministries.

The following are the main objectives of the online kick-off meeting:

- Confirm the project purpose and expected outcome
- Collect the necessary information for assessing Lao PDR's national STI strategic planning and technology foresight capacities
- Share Korean experiences in national strategic planning and technology foresight
- Exchange ideas and opinions on the project schedule and activities

The following are the main activities of the workshop:

- Remarks and introduction of the project by the STEPI project investigator on the project overview including purposes, scope, methodology, implementation, schedule, and results
- Remarks and Q&As by Lao PDR and sharing ideas on the project by the research team in Lao PDR

- Presentations by the Lao experts on Lao's STI policy and strategy, the development and application of technology foresight, and a few cases, to be followed by a lively discussion on key issues and the exchange of ideas and opinions
- Follow-up in-depth meetings with stakeholders to collect their personal views and insights on improving foresight and strategic planning capacities

2.1.2 Team

The research team is responsible for holding the kick-off meeting, conducting interviews with the stakeholders from MOST and other research institutions and decision makers of Lao PDR.

[Table 4-1] Kick-off Meeting

Organization	Name	Project role/Position	Contact
STEPI	Ji Hyun Kim	STEPI, Associate Research Fellow Project Manager (PM)	STI Policy, Technology Foresight
	Deok Soon Yim	STEPI, Senior Research Fellow	Technology Foresight
	Jiyong An	STEPI, Researcher	Project Coordinator

2.2 Kick-off Meeting

2.2.1 List of Kick-off Meeting Participants

[Table 4-2] List of Kick-off Meeting and Field Visit Participants

Name	Office	Position	Contact information
Ji Hyun Kim	STEPI, Associate Research Fellow Project Manager (PM)	STI Policy, Technology Foresight	jihyunkim@stepi.re.kr
Jiyong An	STEPI, Researcher	Project Coordinator	freesoul13@stepi.re.kr
Soumana CHOUNLAMANY	MOST	Director General, Department of Technology and Innovation	Schounlamany@yahoo.com
Keophaivanh DOAUNGSAVANH	MOST	Director General, Cabinet of Academy of Science and Technology	Keophaivanths@yahoo.com
Khamphet VONGDALA	MOST	Former Deputy Minister of Science and Technology	khamphet01@yahoo.com
Phouthasone SINTHAVONG	MOST	Director, Cabinet of Academy of Science and Technology	SoneLao PDR@gmail.com
Viengvilay THONGMANYLA	MOST	Deputy Director, Department of Technology and Innovation	Viengvilayth@gmail.com
Kousim CHANTHAPANYA	MOST	Director, ASEAN Division	kousim@gamil.com
Saykham VOLADET	National Economic Research Institute (NIER)	Acting Director General	Saykhamv@yahoo.com
Sithixay XAYAVONG	National University of Lao PDR	Vice President of Research Ethics	x.sithixay@gmail.com
Sengpasith HOUNGALOUNE	National University of Lao PDR	Lecturer/ Researcher	h.sengpasith@gmail.com

2.2.2 Kick-off Meeting's Main Activities and Agendas

At the kick-off meeting with MOST, the objectives and scope of the Project were discussed. Further specific action plans were discussed, and STEPI and MOST agreed on the main items of the project programs and activities. The key points of the meeting can be summarized as follows:

- Agreed on the Project Action Plan
- Discussed the governance of the project, detailed schedule and activities, and budget
- Emphasized the importance of securing the latest on-site data through local consultants and promised to support Korean experts in connection with the on-site investigation discussed in advance given the COVID-19 situation
- Introduction of other Korean experts who will support the project

[Figure 4-1] Kick-off Meeting



[Figure 4-2] Official Letter for Kick-off Meeting



Lao people's Democratic Republic
Peace Independence Democracy Unity Prosperity

Ministry of Science and Technology
Department of Technology and Innovation

No: **370** /MOST-DTI
Vientiane Capital, Date: **17 JUN 2020**

To: Science and Technology Policy Institute (STEPI)

Subject: Appointment of Focal Points for project "Master Plan for STI Research and Development and Application in Lao PDR"

The Department of Technology and Innovation, MOST, Lao PDR, would like to inform you of the following members as the Focal Points for the project "Master Plan for STI Research and Development and Application in Lao PDR":

1. Ms. Phouthasone SINTHAVON, *Director of Science Policy Division*, Cabinet of Lao Academy of Science
2. Mr. Arthone KHAMMANICHANH, *Deputy Director of ASEAN Division*, Department of Planning and Cooperation
3. Mr. Pasongsay KEOBOUNPHANH, *Deputy Director of High Technology Center*, Department of Technology and Innovation

We sincerely look forward to a successful project cooperation and a greater relationship between our organizations.




Mr. Soumana CHoulamany
Director General
Department of Technology and Innovation

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2.2.2.1 Overview of the Workshop

1) Objective

The workshop aims to enhance Lao PDR's capacity in Science, Technology, and Innovation (STI), especially focusing on the national master plan on STI R&D. The senior-level officials and managers of MOST, relevant ministries, academe, and research institutions will be invited to participate in this online expert workshop. The lectures consist of various training courses from the introduction to the Korean experience, relevant case studies, and methodology to hands-on project exercises.

2) Process

The workshop was discussed in September but has been postponed due to the COVID-19 situation and Lao PDR's legal process and will be held in December 2020.

The online seminars were planned to be held with the aim of transferring cases of Korea's science and technology innovation and national science and technology master plans and methodologies but were postponed due to the re-entry of telecommuting by local ministries (September).

The expected meeting schedule is mid-December 2020. The theme and topic of the workshop are the transfer of Korea's case of science and technology innovation through the presentation of Korean experts, the national science and technology master plan, and the presentation on the current status of science and technology innovation in Lao PDR.

- Collect opinions from local experts at the Lao Ministry of Science and Technology and conduct additional interviews.
- To discuss the project schedule and direction for 2021

It is expected to be an online workshop. Korean experts who will give presentations will be gathered together in a place in Seoul to be connected online with Lao PDR.

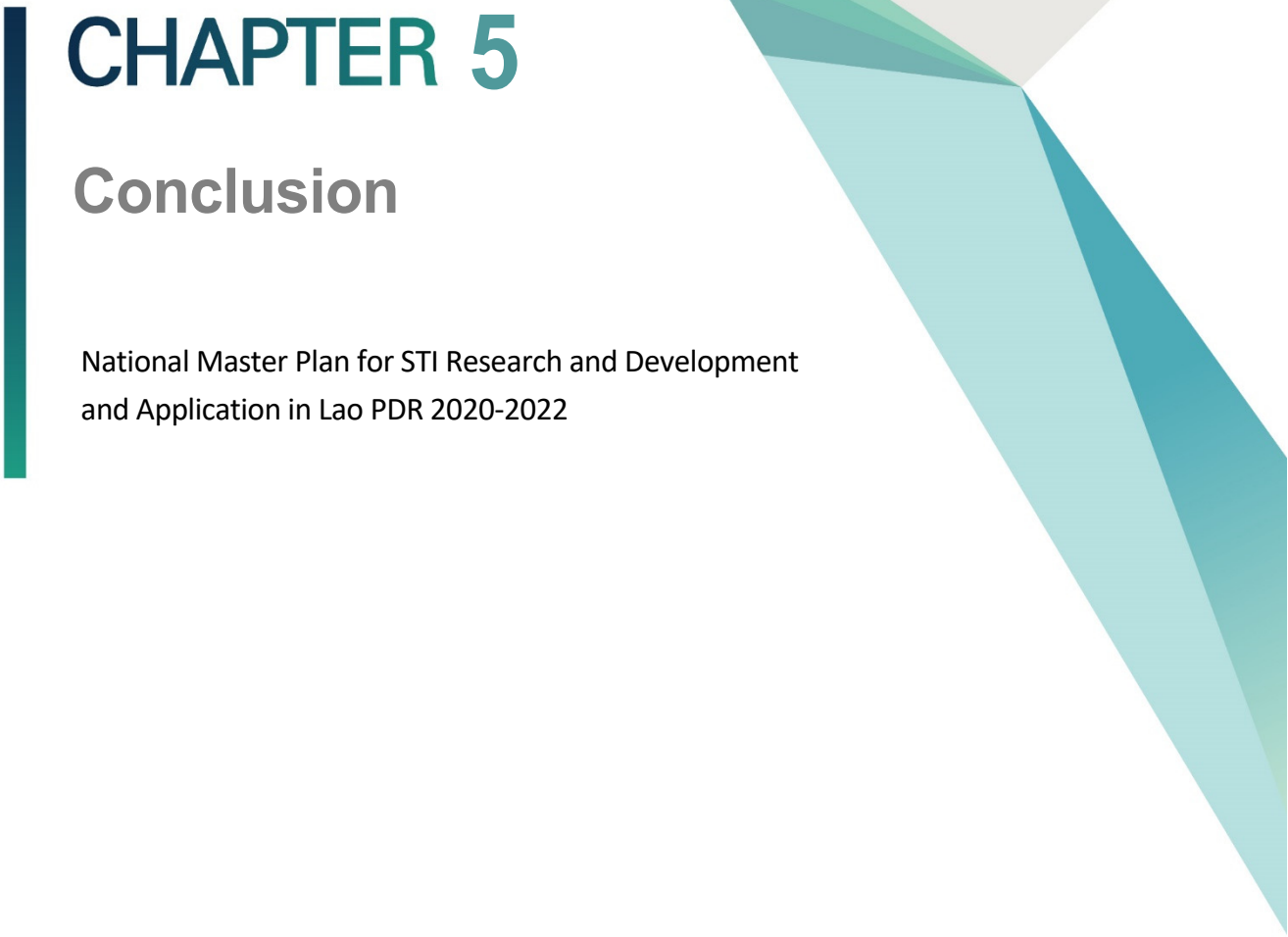
The Expected Participants include 5 Korean domestic experts and about 00 experts from MOST and experts from various fields in Lao PDR.

Revision of budget

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Internal meeting in Lao PDR



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CHAPTER 5

Conclusion

National Master Plan for STI Research and Development
and Application in Lao PDR 2020-2022

Chapter 5. Conclusion

1. STI Policy and System

After gaining its independence from France in 1945, Lao PDR went through the Indochina War and officially won independence through the Geneva Agreement in 1954. Later, with the support of Vietnam, the socialist “Lao People's Democratic Republic” was established in 1975. Lao PDR maintains a one-party dictatorship with the Lao People's Revolutionary Party at the center and a collective leadership system wherein key government posts, including the president, head of state, prime minister, and speaker of the National Assembly concurrently serve as the top political bureau in charge of decision making of the party's central committee.

With the goal of formulating a master plan enhancing support for the development of the 8th National Socio-Economic Development Plan and contributing to the preparation of the 9th National Socio-Economic Development Plan, Lao PDR's Development Strategy and Vision up to year 2030 has identified the National Science and Technology Development Program as a driving force for economic growth. To be sustainable, green, and consistent with the SMEs Development Strategies, the national development plan needs to be associated with the transformation of SMEs for the old business production going to the new modern economics of science, technology and innovation.

The objective of the K-Innovation ODA Project with Lao PDR (2020-2022) is to set up the first National Master Plan for STI R&D and Application in Lao PDR and Master Plan on the Application of Technology and Innovation for Sectoral Support for National Social-Economic Development Growth in the Future as well as Sectors under development. Moreover, the

project aims to strengthen the capacity of key institutions of the Ministry of Science and Technology in the mid- and long-term strategic planning of STI in Lao PDR.

It is necessary to make the government establish and implement an STI strategy systematically with active commitment to making STI contribute to national socio-economic development. However, Lao PDR does not have much experience in making and implementing such STI policies linked with the national development plan.

There is also a need to consider improving the R&D structure to increase the performance of the R&D system given the limited R&D resources. The role of R&D organizations such as KIST, which is a government/public research institute, should be established. In addition, the linkage system with universities and industries needs to be strengthened for these public research institutes to create R&D synergy effects.

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Based on the confirmation and acceptance of the project proposal submitted by MOST, the Korean government affirmed support for Lao PDR in making faster progress toward

achieving Lao PDR's STI development goals through the K-Innovation ODA program "National Master Plan for STI Research and Development and Application in Lao PDR 2020-2022."

The scope of the Project has been defined based on Lao PDR's STI policy priorities and the lessons learned from the Korean experiences in national STI policies and technology foresight. Accordingly, the Project framework includes the following: i) review of Lao PDR's national STI strategies and technology foresight experiences; ii) research on the Korean experiences in technology foresight and national STI strategies; iii) local workshops for the diagnosis of Lao PDR's technology foresight capacities and for the consensus on the policy consultation's direction and contents; iv) capacity-building workshop with technical training programs to enhance the technology foresight capacity of Lao government officials and experts; and v) development of implementation plan for technology foresight by VISTI and review process by STEPI and Korean experts.

At the kick-off meeting with MOST, the objectives and scope of the Project were discussed. STEPI and MOST discussed further specific action plans and agreed on the project programs and activities. The research team suggested implications and recommendations on how to develop Lao PDR's capacity for national STI strategy. Regarding the national STI strategy, it is necessary to make the government establish and implement an STI strategy systematically with active commitment to making STI contribute to national socio-economic development. Lao PDR needs to promote strategic investments and systematic management to achieve the best possible outcomes given the limited R&D resources. It also needs to improve its R&D structure to increase the performance of the R&D system considering the limited R&D resources. For this, the role of R&D organizations should be clearly identified.

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